



# **ene6K5830**

## **8051 Flash MCU with 24 Channel PWM/LED driver**

### **Datasheet**

Revision 1.1

Aug. 2018

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### Revision

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|----------|-----------------------|---------|
| 1.0      | Initial release       | 2017/10 |
| 1.1      | Updated SMBus Timming | 2018/08 |

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## 1. General Description

### 1.1 Overview

ene6K5830 is a 8-bit flash microcontroller embedded with 8051. The operating voltage is from 2.7V to 5.5V and operating temperature is from -40°C to 85°C. ene6K5830 include 28 MHz high-accuracy and high speed oscillators which require no external components for system frequency. ene6K5830 support: 28K bytes flash memory, 4K bytes flash with boot ROM, 256bytes IRAM and 2048 bytes XRAM, 24×10-bit PWM with high sink current 20mA\*15 and 50mA\*9, I<sup>2</sup>C and SPI and UART communication interfaces.

ene6K5830 will find excellent use in a huge range of modern LED product applications such as VGA card or Motherboard LED dimming.

### 1.2 IC Features

- **Core**
  - 2T 8051 MCU
  - maximum system frequency: 28 MHz
  - operation voltage: 2.7V ~ 5.5V
  - operating temperature range: -40°C - 85°C
  - embedded RC high-accuracy oscillator at 28MHz ± 3% -40°C ~ 85°C that can save external
- **Memory**
  - 28K bytes flash memory
  - 4K bytes flash with boot ROM
  - 256 bytes IRAM
  - 2K bytes XRAM
  - online system update ISP (In-System Programming)
  - online circuit update ICP (In-Circuit Programming)
- **Maximum of 30 general GPIOs with 5V Torrance**
- **24×10-bit PWM with high sink output**
- **Connectivity**
  - 1 set of SMBus/I<sup>2</sup>C device (maximum 400 kbps) (Master or Slave by F/W)
  - 1 set of UART interface
  - 1 set of SMBus Slave interface
  - 1 set of SPI interface (maximum 16 Mbps)
- **2 units of 16-bit timer/counter**
- **LVR resetting at low voltage**
- **LVD detection at low voltage**
- **16 bit watch-dog timer**

### 1.3 Selection Guide

| Part NO.     | Flash ROM | RAM | I/O | PWM      | SMBus Master/ Slave | UART / SPI | Package |
|--------------|-----------|-----|-----|----------|---------------------|------------|---------|
| ene6K5830UA0 | 28K       | 2K  | 30  | 24CHx10b | 1/2                 | 1/1        | QFN32   |

Table1.1 selection guide

○ VDD=2.7~5.5V; LVD=2.5~3.2V; IRAM= 256B; B=bytes; b=bit; CH= channel

### 1.4 System block diagram

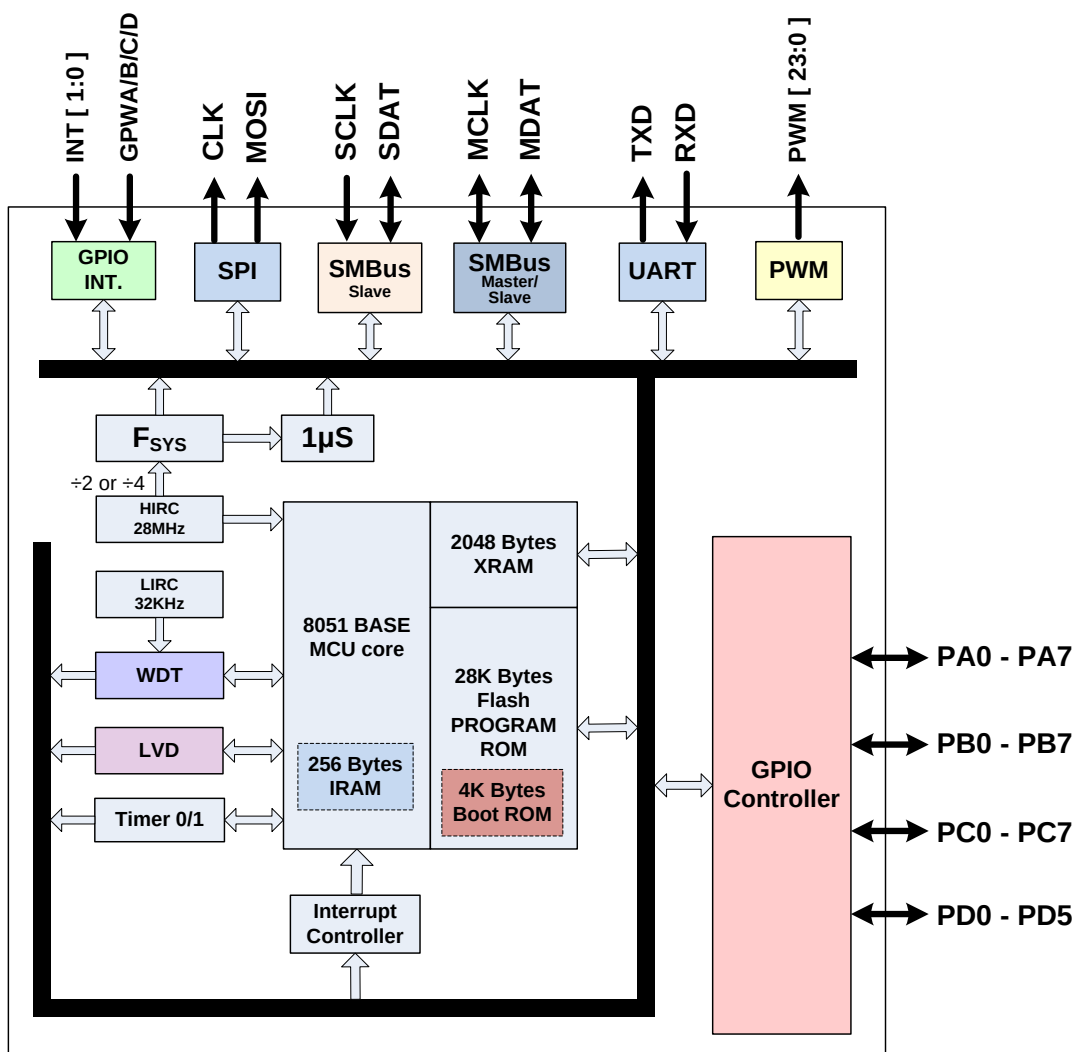


Fig1.1 ene6K5830 System block diagram

## 1.5 Pin Assignment

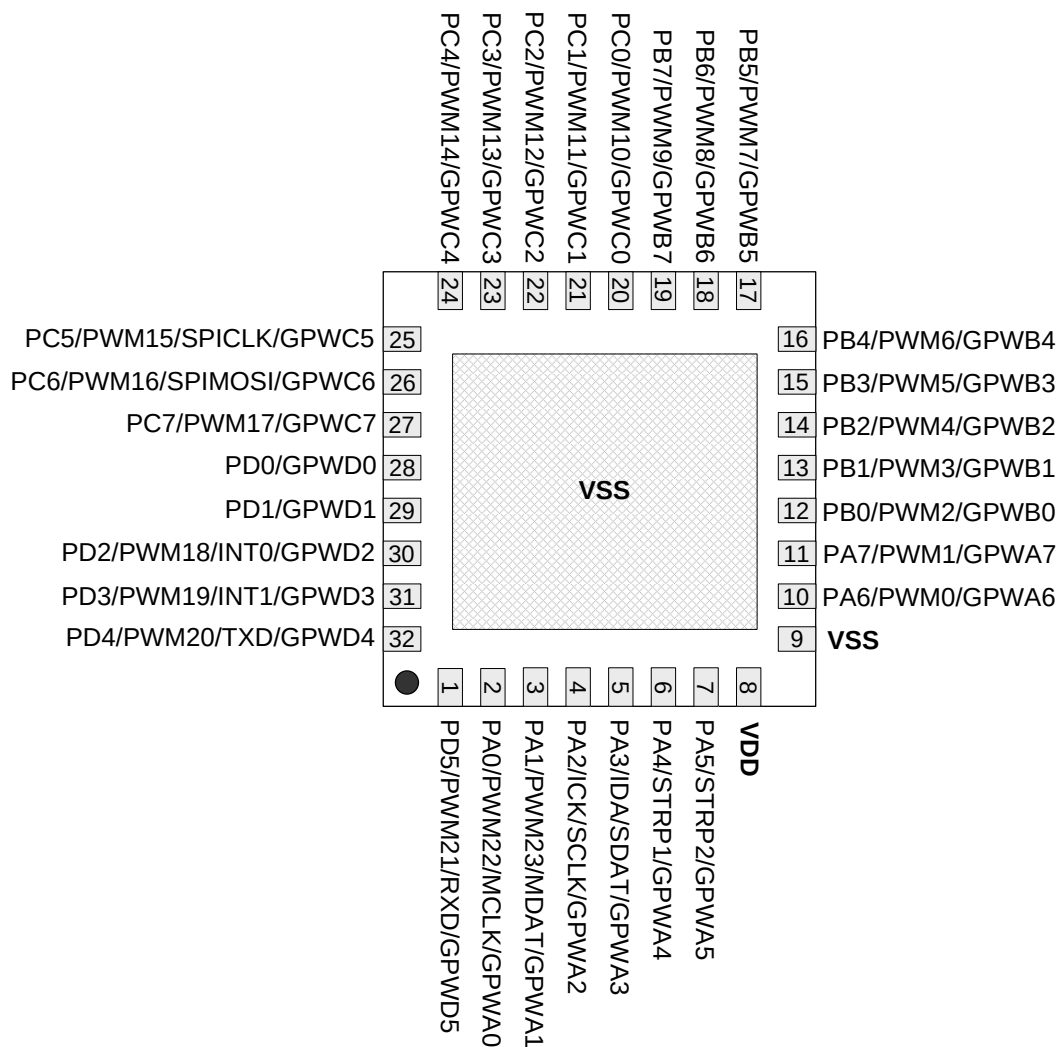


Fig1.2 ene6K5830 QFN32 Pin Assignment

### ➤ SMBus Address setting

| Pin6, Pin7    | (0, 0) | (0, 1) | (1, 0) | (1, 1) |
|---------------|--------|--------|--------|--------|
| SMBus Address | 0xCE   | 0xCC   | 0x9E   | 0x9C   |

## 1.6 Pinout Description

| Pin Name                 | Pin multiplexing | Types | Reset state | Functional summary                                              | Note |
|--------------------------|------------------|-------|-------------|-----------------------------------------------------------------|------|
| PA0/PWM22/MCLK/<br>GPWA0 | PA0              | I/O   | HiZ         | PA0                                                             |      |
|                          | PWM22            | O     | --          | 10 bit PWM Channel output 22                                    |      |
|                          | MCLK             | I/O   | --          | SMBus clock<br>Master mode : clock out<br>Slave mode : clock in |      |
|                          | GPWA0            | I     | --          | External wake-up interrupt input                                |      |
| PA1/PWM23/MDAT/<br>GPWA1 | PA1              | I/O   | HiZ         | PA1                                                             |      |
|                          | PWM23            | O     | --          | 10 bit PWM Channel output 23                                    |      |
|                          | MDAT             | I/O   | --          | SMBus master/slave data                                         |      |
|                          | GPWA1            | I     | --          | External wake-up interrupt input                                |      |
| PA2/ICK/SCLK/ GPWA2      | PA2              | I/O   | --          | PA2                                                             |      |
|                          | ICK              | I     | --          | On chip debugger function clock pin                             |      |
|                          | SCLK             | I     | HiZ         | SMBus slave clock input and<br>programmable clock input         |      |
|                          | GPWA2            | I     | --          | External wake-up interrupt input                                |      |
| PA3/IDA/SDAT/ GPWA3      | PA3              | I/O   | --          | PA3                                                             |      |
|                          | IDA              | I/O   | --          | On chip debugger function data pin                              |      |
|                          | SDAT             | I/O   | HiZ         | SMBus slave data and programmable data<br>input or output       |      |
|                          | GPWA3            | I     | --          | External wake-up interrupt input                                |      |
| PA4/Strp1/GPWA4          | PA4              | I/O   | --          | PA4                                                             |      |
|                          | Strp1            | I     | Pull Hi     | SMBus Address setting                                           |      |
|                          | GPWA4            | I     | --          | External wake-up interrupt input                                |      |
| PA5/Strp1/GPWA5          | PA5              | I/O   | --          | PA5                                                             |      |
|                          | Strp2            | I     | Pull Hi     | SMBus Address setting                                           |      |
|                          | GPWA5            | I     | --          | External wake-up interrupt input                                |      |
| PA6/PWM0/GPWA6           | PA6              | I/O   | HiZ         | PA6                                                             |      |
|                          | PWM0             | O     | --          | 10 bit PWM Channel output 0 with 50mA<br>sink current           |      |
|                          | GPWA6            | I     | --          | External wake-up interrupt input                                |      |
| PA7/PWM1/GPWA7           | PA7              | I/O   | HiZ         | PA6                                                             |      |
|                          | PWM1             | O     | --          | 10 bit PWM Channel output 1 with 50mA<br>sink current           |      |
|                          | GPWA7            | I     | --          | External wake-up interrupt input                                |      |
| PB0/PWM2/GPWB0           | PB0              | I/O   | HiZ         | PB0                                                             |      |
|                          | PWM2             | O     | --          | 10 bit PWM Channel output 2 with 50mA<br>sink current           |      |
|                          | GPWB0            | I     | --          | External wake-up interrupt input                                |      |

(Continuance 1)

| Pin Name        | Pin multiplexing | Types | Reset state | Functional summary                                 | Note |
|-----------------|------------------|-------|-------------|----------------------------------------------------|------|
| PB1/PWM3/GPWB1  | PB1              | I/O   | HiZ         | PB1                                                |      |
|                 | PWM3             | O     | --          | 10 bit PWM Channel output 3 with 50mA sink current |      |
|                 | GPWB1            | I     | --          | External wake-up interrupt input                   |      |
| PB2/PWM4/GPWB2  | PB2              | I/O   | HiZ         | PB2                                                |      |
|                 | PWM4             | O     | --          | 10 bit PWM Channel output 4 with 50mA sink current |      |
|                 | GPWB2            | I     | --          | External wake-up interrupt input                   |      |
| PB3/PWM5/GPWB3  | PB3              | I/O   | HiZ         | PB3                                                |      |
|                 | PWM5             | O     | --          | 10 bit PWM Channel output 5 with 50mA sink current |      |
|                 | GPWB3            | I     | --          | External wake-up interrupt input                   |      |
| PB4/PWM6/GPWB4  | PB4              | I/O   | HiZ         | PB2                                                |      |
|                 | PWM6             | O     | --          | 10 bit PWM Channel output 6 with 50mA sink current |      |
|                 | GPWB4            | I     | --          | External wake-up interrupt input                   |      |
| PB5/PWM7/GPWB5  | PB5              | I/O   | HiZ         | PB5                                                |      |
|                 | PWM7             | O     | --          | 10 bit PWM Channel output 7 with 50mA sink current |      |
|                 | GPWB5            | I     | --          | External wake-up interrupt input                   |      |
| PB6/PWM8/GPWB6  | PB6              | I/O   | HiZ         | PB6                                                |      |
|                 | PWM8             | O     | --          | 10 bit PWM Channel output 8 with 50mA sink current |      |
|                 | GPWB6            | I     | --          | External wake-up interrupt input                   |      |
| PB7/PWM9/GPWB7  | PB7              | I/O   | HiZ         | PB7                                                |      |
|                 | PWM9             | O     | --          | 10 bit PWM Channel output 9                        |      |
|                 | GPWB7            | I     | --          | External wake-up interrupt input                   |      |
| PC0/PWM10/GPWC0 | PC0              | I/O   | HiZ         | PC0                                                |      |
|                 | PWM10            | O     | --          | 10 bit PWM Channel output 10                       |      |
|                 | GPWC0            | I     | --          | External wake-up interrupt input                   |      |
| PC1/PWM11/GPWC1 | PC1              | I/O   | HiZ         | PC1                                                |      |
|                 | PWM11            | O     | --          | 10 bit PWM Channel output 11                       |      |
|                 | GPWC1            | I     | --          | External wake-up interrupt input                   |      |
| PC2/PWM12/GPWC2 | PC2              | I/O   | HiZ         | PC2                                                |      |
|                 | PWM12            | O     | --          | 10 bit PWM Channel output 12                       |      |
|                 | GPWC2            | I     | --          | External wake-up interrupt input                   |      |

(Continuance 2)

| Pin Name                    | Pin multiplexing | Types | Reset state | Functional summary                | Note |
|-----------------------------|------------------|-------|-------------|-----------------------------------|------|
| PC3/PWM13/GPWC3             | PC3              | I/O   | HiZ         | PC3                               |      |
|                             | PWM10            | O     | --          | 10 bit PWM Channel output 13      |      |
|                             | GPWC3            | I     | --          | External wake-up interrupt input  |      |
| PC4/PWM14/GPWC4             | PC4              | I/O   | HiZ         | PC4                               |      |
|                             | PWM14            | O     | --          | 10 bit PWM Channel output 14      |      |
|                             | GPWC4            | I     | --          | External wake-up interrupt input  |      |
| PC5/PWM15/SPICLK/<br>GPWC5  | PC5              | I/O   | HiZ         | PC5                               |      |
|                             | PWM15            | O     | --          | 10 bit PWM Channel output 15      |      |
|                             | SPICLK           | O     | --          | SPI Clock output                  |      |
|                             | GPWC5            | I     | --          | External wake-up interrupt input  |      |
| PC6/PWM16/SPIMOSI/<br>GPWC6 | PC6              | I/O   | HiZ         | PC6                               |      |
|                             | PWM16            | O     | --          | 10 bit PWM Channel output 15      |      |
|                             | SPIMOSI          | O     | --          | SPI data path master out slave in |      |
|                             | GPWC5            | I     | --          | External wake-up interrupt input  |      |
| PC7/PWM17/GPWC7             | PC7              | I/O   | HiZ         | PC7                               |      |
|                             | PWM17            | O     | --          | 10 bit PWM Channel output 17      |      |
|                             | GPWC7            | I     | --          | External wake-up interrupt input  |      |
| PD0/GPWD0                   | PD0              | I/O   | HiZ         | PD0                               |      |
|                             | GPWD0            | I     | --          | External wake-up interrupt input  |      |
| PD1/GPWD1                   | PD1              | I/O   | HiZ         | PD1                               |      |
|                             | GPWD1            | I     | --          | External wake-up interrupt input  |      |
| PD2/PWM18/INT0/<br>GPWD2    | PD2              | I/O   | HiZ         | PD2                               |      |
|                             | PWM18            | O     | --          | 10 bit PWM Channel output 18      |      |
|                             | INT0             | I     | --          | External interrupt input 0        |      |
|                             | GPWD2            | I     | --          | External wake-up interrupt input  |      |
| PD3/PWM19/INT1/<br>GPWD3    | PD3              | I/O   | HiZ         | PD3                               |      |
|                             | PWM19            | O     | --          | 10 bit PWM Channel output 19      |      |
|                             | INT0             | I     | --          | External interrupt input 0        |      |
|                             | GPWD3            | I     | --          | External wake-up interrupt input  |      |
| PD4/PWM20/TXD/<br>GPWD4     | PD4              | I/O   | --          | PD4                               |      |
|                             | PWM20            | O     | --          | 10 bit PWM Channel output 20      |      |
|                             | TXD              | O     | --          | UART TX ports                     |      |
|                             | GPWD4            | I     | --          | External wake-up interrupt input  |      |

(Continuance 3)

| Pin Name                | Pin multiplexing | Types | Reset state | Functional summary                  | Note |
|-------------------------|------------------|-------|-------------|-------------------------------------|------|
| PD5/PWM21/RXD/<br>GPWD5 | PD5              | I/O   | --          | PD5                                 |      |
|                         | PWM21            | O     | --          | 10 bit PWM Channel output 21        |      |
|                         | RXD              | I     | --          | UART RX ports                       |      |
|                         | GPWC3            | I     | --          | External wake-up interrupt input    |      |
| VDD                     | VDD              | P     | --          | Chip power (2.7V-5.5V)              |      |
| VSS                     | VSS              | P     | --          | Ground, reference point for Chip 0V |      |

➤ Note 1: I/O=Input/Output, O=Output, I=Input, P=Power, HiZ=High impedance.

## 2. Registers Mapping and Description

### 2.1 Misc. Register File

base address : 0xF000

| offset | register abbr. | register name                      |      |                                                                                                                                                                                                                | pwr-on value |
|--------|----------------|------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                | attr | description                                                                                                                                                                                                    |              |
| 0x00   | MISCPMU        | <b>PMU configuration</b>           |      |                                                                                                                                                                                                                | 0x01         |
|        |                | 7-5                                | NA   | <b>Reserved</b>                                                                                                                                                                                                |              |
|        |                | 4                                  | RW   | <b>enable fast stand-by mode</b><br>(macro_stby = ~pmu_clk_run)                                                                                                                                                |              |
|        |                | 3                                  | RW   | <b>OSC32M control in low power mode</b><br>0: OSC32M disabled in low power mode<br>1: OSC32M enabled in low power mode                                                                                         |              |
|        |                | 2                                  | WO   | <b>enter STOP mode</b><br>writing this bit = 1, as well as E51 PCON STOP<br>note:<br>1. if wdt_clk != external OSC, don't enter stop mode<br>2. if I/F is busy, this bit is invalid, E51 PCON STOP is the same |              |
|        |                | 1                                  | RW   | <b>enable WDT interrupt wakeup</b>                                                                                                                                                                             |              |
|        |                | 0                                  | RW   | <b>enable GPW wakeup</b>                                                                                                                                                                                       |              |
| 0x01   | TXDMUX         | <b>TX data output path for E51</b> |      |                                                                                                                                                                                                                | 0x1C         |
|        |                | 7-5                                | NA   | <b>Reserved</b>                                                                                                                                                                                                |              |
|        |                | 4-0                                | RW   | <b>select a GPIO for TX data output</b><br>0x0~0x07: PA0~PA7<br>0x08~0x0F: PB0~PB7<br>0x10~0x17: PC0~PC7<br>0x18~0x1D: PD0~PD5<br>others: reserved<br>this is effective only when TX data output enabled       |              |
| 0x02   | SMB0Addr       | <b>SMBus0 Slave device address</b> |      |                                                                                                                                                                                                                | 0xCE         |
|        |                | 7-1                                | RW   | <b>device address</b><br>PA[5:4]=2'h0 : 7'h4E (0x9C)<br>PA[5:4]=2'h1 : 7'h4F (0x9E)<br>PA[5:4]=2'h2 : 7'h66 (0xCC)<br>PA[5:4]=2'h3 : 7'h67 (0xCE)                                                              |              |
|        |                | 0                                  | RO   | <b>direction (host-initiated)</b><br>0/1: write/read                                                                                                                                                           |              |
| 0x03   | SMB0CTRL0      | <b>SMBus0 slave control0</b>       |      |                                                                                                                                                                                                                | 0x15         |
|        |                | 7                                  | RW   | <b>delay SDA input falling edge</b>                                                                                                                                                                            |              |
|        |                | 6                                  | RW   | <b>delay SDA input rising edge</b>                                                                                                                                                                             |              |
|        |                | 5                                  | NA   | <b>Reserved</b>                                                                                                                                                                                                |              |
|        |                | 4                                  | RW   | <b>SMB0 slave enable</b>                                                                                                                                                                                       |              |
|        |                | 3                                  | NA   | <b>Reserved</b>                                                                                                                                                                                                |              |
|        |                | 2-0                                | RW   | <b>turn-around cycle</b>                                                                                                                                                                                       |              |
| 0x04   | SMB0CTRL1      | <b>SMBus0 slave control1</b>       |      |                                                                                                                                                                                                                | 0x00         |
|        |                | 7-1                                | NA   | <b>Reserved</b>                                                                                                                                                                                                |              |
|        |                | 0                                  | RW   | <b>drive SMB DAT 1T high after completion of driving low enable</b>                                                                                                                                            |              |
| 0x05   | MISC01         | <b>MISC01</b>                      |      |                                                                                                                                                                                                                | 0x80         |
|        |                | 7                                  | RW   | <b>RST# 40KOhm pull-up pull-up enable</b>                                                                                                                                                                      |              |
|        |                | 6                                  | NA   | <b>Reserved</b>                                                                                                                                                                                                |              |
|        |                | 5                                  | RW   | <b>SPI enable</b>                                                                                                                                                                                              |              |
|        |                | 4-0                                | NA   | <b>Reserved</b>                                                                                                                                                                                                |              |

| offset | register abbr. | register name                                |      |                                                                                                                                                                                                   | pwr-on value |
|--------|----------------|----------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                          | attr | description                                                                                                                                                                                       |              |
| 0x06   | SMB0CTRL2      | <b>SMBus0 slave control2</b>                 |      |                                                                                                                                                                                                   | 0x0F         |
|        |                | 7                                            | RW   | delay SCL input falling edge                                                                                                                                                                      |              |
|        |                | 6                                            | RW   | delay SCL input rising edge                                                                                                                                                                       |              |
|        |                | 5-0                                          | NA   | Reserved                                                                                                                                                                                          |              |
| 0x07   | LVDCTL         | <b>LVD control / bonding option control</b>  |      |                                                                                                                                                                                                   | 0xA0         |
|        |                | 7-5                                          | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 4                                            | RW   | LVDA enable                                                                                                                                                                                       |              |
|        |                | 3-1                                          | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 0                                            | RW   | LVDA interrupt enable                                                                                                                                                                             |              |
| 0x08   | LVDSTA         | <b>LVD status</b>                            |      |                                                                                                                                                                                                   | 0x10         |
|        |                | 7-5                                          | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 4                                            | RO   | LVDA macro output status<br>1: VCC>Vth, 0: VCC< Vth                                                                                                                                               |              |
|        |                | 3-1                                          | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 0                                            | W1C  | LVDA pending flag (detect falling edge)                                                                                                                                                           |              |
| 0x09   | MISCCLK        | <b>clock select / bonding option control</b> |      |                                                                                                                                                                                                   | 0x00         |
|        |                | 7-                                           | RW   | DATA RAM clock gator enable                                                                                                                                                                       |              |
|        |                | 6                                            | RW   | E51 clock gator enable                                                                                                                                                                            |              |
|        |                | 5-4                                          | RW   | 1us pulse tuning<br>0/1/2/3: 0/-1/+1/+2<br>(unit: 14MHz clock)<br>if main clock is 1MHz, '-1' option is meaningless                                                                               |              |
|        |                | 3                                            | RW   | XBI clock gator enable                                                                                                                                                                            |              |
|        |                | 2-0                                          | RW   | main clock divisor<br>0/1/2/3/4: 4/8/16/32/2 (8/4/Rev./Rev./16 MHz)                                                                                                                               |              |
| 0x0D   | RXDMUX         | <b>RX data input path for E51</b>            |      |                                                                                                                                                                                                   | 0x1D         |
|        |                | 7-5                                          | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 4-0                                          | RW   | select a GPIO for RX data output<br>0x0~0x07: PA0~PA7<br>0x08~0x0F: PB0~PB7<br>0x10~0x17: PC0~PC7<br>0x18~0x1D: PD0~PD5<br>others: reserved<br>this is effective only when RX data output enabled |              |
| 0x10   | E51MISC        | <b>E51 configuration</b>                     |      |                                                                                                                                                                                                   | 0x04         |
|        |                | 7-6                                          | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 5                                            | RW   | e51 txclk enable                                                                                                                                                                                  |              |
|        |                | 4                                            | RW   | e51 tx enable                                                                                                                                                                                     |              |
|        |                | 3                                            | RW   | reserved                                                                                                                                                                                          |              |
|        |                | 2                                            | RW   | WDT reset control<br>0/1: reset all function/E51 only                                                                                                                                             |              |
|        |                | 1                                            | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 0                                            | RW   | E51 software reset<br>0/1: run/reset and stop                                                                                                                                                     |              |
| 0x1B   | TRAPSTA        | <b>HW trapping status</b>                    |      |                                                                                                                                                                                                   | 0x--         |
|        |                | 7                                            | RO   | POR status                                                                                                                                                                                        |              |
|        |                | 6-5                                          | RO   | SMB IF Address Trap (PA[5:4])                                                                                                                                                                     |              |
|        |                | 4                                            | NA   | Reserved                                                                                                                                                                                          |              |
|        |                | 3                                            | RO   | Bypass trap waiting time                                                                                                                                                                          |              |
|        |                | 2                                            | RO   | Flash bist mode                                                                                                                                                                                   |              |
|        |                | 1                                            | RO   | Stop 8051, ISP Only(0x55)                                                                                                                                                                         |              |
|        |                | 0                                            | NA   | Reserved                                                                                                                                                                                          |              |

| offset | register abbr. | register name              |      |                                                                                                                                                             | pwr-on value |
|--------|----------------|----------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                        | attr | description                                                                                                                                                 |              |
| 0x42   | TRAP_SEL       | <b>Trap Mode Select</b>    |      |                                                                                                                                                             | 0x00         |
|        |                | 7-0                        | R/W  | To decide which trap mode.<br>55: Stop 8051<br>AA: Clock Test Mode<br>5A: Flash BIST Mode<br>A5: Bypass Mode<br>B4: EEPROM BIST Mode<br>Others: Normal Mode |              |
| 0x7B   | VIHL_CTRL      | <b>VIH/VIL Control</b>     |      |                                                                                                                                                             | 0x00         |
|        |                | 7-2                        | NA   | <b>Reserved</b>                                                                                                                                             |              |
|        |                | 1                          | R/W  | VIH/VIL of 3.3V select<br>0: VIH/VIL of VCC , 1: VIH/VIL of 3.3V                                                                                            |              |
|        |                | 0                          | R/W  | VIH/VIL latch control<br>0: Continuously latch, 1: Stop latching                                                                                            |              |
| 0x7C   | SMB1CTRL1      | <b>SMB1 slave control1</b> |      |                                                                                                                                                             | 0x05         |
|        |                | 7                          | R/W  | delay SDA1 input falling edge                                                                                                                               |              |
|        |                | 6                          | R/W  | delay SDA1 input rising edge                                                                                                                                |              |
|        |                | 5                          | NA   | <b>Reserved</b>                                                                                                                                             |              |
|        |                | 4                          | R/W  | SMB1 slave enable                                                                                                                                           |              |
|        |                | 3                          | NA   | <b>Reserved</b>                                                                                                                                             |              |
|        |                | 2-0                        | R/W  | SMB1 turn-around cycle                                                                                                                                      |              |
| 0x7D   | SMB1CTRL2      | <b>SMB1 slave control2</b> |      |                                                                                                                                                             | 0x00         |
|        |                | 7-3                        | NA   | <b>Reserved</b>                                                                                                                                             |              |
|        |                | 2                          | R/W  | drive SMB1 DAT 1T high after completion of driving low enable<br>0/1: disable/enable                                                                        |              |
|        |                | 1                          | R/W  | delay SCL1 input falling edge                                                                                                                               |              |
|        |                | 0                          | R/W  | delay SCL1 input rising edge                                                                                                                                |              |
| 0x7E   | SMB1ADDR       | <b>SMB1 device address</b> |      |                                                                                                                                                             | 0x00         |
|        |                | 7-1                        | R/W  | device address                                                                                                                                              |              |
|        |                | 0                          | RO   | direction (host-initiated)<br>0/1: write/read                                                                                                               |              |

## 2.2 WDT

base address : 0xF100

| offset | register abbr. | register name            |      |                                                                                                              | pwr-on value |
|--------|----------------|--------------------------|------|--------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                      | attr | description                                                                                                  |              |
| 0x00   | WDTCFG         | <b>WDT configuration</b> |      |                                                                                                              | 0x00         |
|        |                | 7-4                      | NA   | Reserved                                                                                                     |              |
|        |                | 3                        | RW   | timer enable                                                                                                 |              |
|        |                | 2                        | NA   | Reserved                                                                                                     |              |
|        |                | 1                        | RW   | interrupt enable                                                                                             |              |
|        |                | 0                        | WO   | timer reset<br>WDT and WDTPF[1:0] will be reset.                                                             |              |
| 0x01   | WDTPF          | <b>WDT interrupt</b>     |      |                                                                                                              | 0x00         |
|        |                | 7-2                      | NA   | Reserved                                                                                                     |              |
|        |                | 1                        | W1C  | WDT semi-timeout pending flag                                                                                |              |
|        |                | 0                        | W1C  | WDT reset pending flag<br>WDT reset asserts if timer is timeout and WDTPF[1] is true.                        |              |
| 0x02   | WDTVAL         | <b>WDT counter value</b> |      |                                                                                                              | 0x00         |
|        |                | 7-0                      | RW   | Number of clock cycles before WDT reset occurs<br>0: reserved<br>1: 256<br>2: 256*2<br>.....<br>255: 256*255 |              |
| 0x03   | WDTCLK         | <b>WDT Timer</b>         |      |                                                                                                              | 0x00         |
|        |                | 7-0                      | RO   | WDT internal Timer value                                                                                     |              |

## 2.3 GPIO

base address : 0xF200

| offset | register abbr. | register name                                     |      |                                       | pwr-on value |
|--------|----------------|---------------------------------------------------|------|---------------------------------------|--------------|
|        |                | bit                                               | attr | description                           |              |
| 0x00   | GPIOMD0        | <b>GPIO PA6, PA7, PB0~PB5 mode select</b>         |      |                                       | 0x00         |
|        |                | 7                                                 | R/W  | PB5 is GPIO or PWM7<br>0/1: GPIO/PWM  |              |
|        |                | 6                                                 | R/W  | PB4 is GPIO or PWM6<br>0/1: GPIO/PWM  |              |
|        |                | 5                                                 | R/W  | PB3 is GPIO or PWM5<br>0/1: GPIO/PWM  |              |
|        |                | 4                                                 | R/W  | PB2 is GPIO or PWM4<br>0/1: GPIO/PWM  |              |
|        |                | 3                                                 | R/W  | PB1 is GPIO or PWM3<br>0/1: GPIO/PWM  |              |
|        |                | 2                                                 | R/W  | PB0 is GPIO or PWM2<br>0/1: GPIO/PWM  |              |
|        |                | 1                                                 | R/W  | PA7 is GPIO or PWM1<br>0/1: GPIO/PWM  |              |
|        |                | 0                                                 | R/W  | PA6 is GPIO or PWM0<br>0/1: GPIO/PWM  |              |
| 0x01   | GPIOMD1        | <b>GPIO PB6~PB7, PC0~PC5 mode select</b>          |      |                                       | 0x00         |
|        |                | 7                                                 | R/W  | PC5 is GPIO or PWM15<br>0/1: GPIO/PWM |              |
|        |                | 6                                                 | R/W  | PC4 is GPIO or PWM14<br>0/1: GPIO/PWM |              |
|        |                | 5                                                 | R/W  | PC3 is GPIO or PWM13<br>0/1: GPIO/PWM |              |
|        |                | 4                                                 | R/W  | PC2 is GPIO or PWM12<br>0/1: GPIO/PWM |              |
|        |                | 3                                                 | R/W  | PC1 is GPIO or PWM11<br>0/1: GPIO/PWM |              |
|        |                | 2                                                 | R/W  | PC0 is GPIO or PWM10<br>0/1: GPIO/PWM |              |
|        |                | 1                                                 | R/W  | PB7 is GPIO or PWM9<br>0/1: GPIO/PWM  |              |
|        |                | 0                                                 | R/W  | PB6 is GPIO or PWM8<br>0/1: GPIO/PWM  |              |
| 0x02   | GPIOMD2        | <b>GPIO PC6~PC7, PD0~PD5, PA0~PA1 mode select</b> |      |                                       | 0x00         |
|        |                | 7                                                 | R/W  | PA1 is GPIO or PWM23<br>0/1: GPIO/PWM |              |
|        |                | 6                                                 | R/W  | PA0 is GPIO or PWM22<br>0/1: GPIO/PWM |              |
|        |                | 5                                                 | R/W  | PD5 is GPIO or PWM21<br>0/1: GPIO/PWM |              |
|        |                | 4                                                 | R/W  | PD4 is GPIO or PWM20<br>0/1: GPIO/PWM |              |
|        |                | 3                                                 | R/W  | PD3 is GPIO or PWM19<br>0/1: GPIO/PWM |              |
|        |                | 2                                                 | R/W  | PD2 is GPIO or PWM18<br>0/1: GPIO/PWM |              |
|        |                | 1                                                 | R/W  | PC7 is GPIO or PWM17<br>0/1: GPIO/PWM |              |
|        |                | 0                                                 | R/W  | PC6 is GPIO or PWM16<br>0/1: GPIO/PWM |              |

| offset | register abbr. | register name                                |      |                                                                                                                | pwr-on value |
|--------|----------------|----------------------------------------------|------|----------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                          | attr | description                                                                                                    |              |
| 0x03   | GPIOAOE        | <b>GPIO PA0~PA7 output enable</b>            |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | for all of these OE registers, this output enable register is not effective in PWM mode<br>0/1: disable/enable |              |
| 0x04   | GPIOBOE        | <b>GPIO PB0~PB7 output enable</b>            |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | for all of these OE registers, this output enable register is not effective in PWM mode<br>0/1: disable/enable |              |
| 0x05   | GPIOCOE        | <b>GPIO PC0~PC7 output enable</b>            |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | for all of these OE registers, this output enable register is not effective in PWM mode<br>0/1: disable/enable |              |
| 0x06   | GPIODOE        | <b>GPIO PD0~PD5 output enable</b>            |      |                                                                                                                | 0x00         |
|        |                | 7-6                                          | NA   | <b>Reserved</b>                                                                                                |              |
|        |                | 5-0                                          | R/W  | for all of these OE registers, this output enable register is not effective in PWM mode<br>0/1: disable/enable |              |
| 0x07   | GPIOADO        | <b>GPIO PA0~PA7 output data</b>              |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | <b>PA0~PA7 output data</b>                                                                                     |              |
| 0x08   | GPIOBDO        | <b>GPIO PB0~PB7 output data</b>              |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | <b>PB0~PB7 output data</b>                                                                                     |              |
| 0x09   | GPIOCDO        | <b>GPIO PC0~PC7 output data</b>              |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | <b>PC0~PC7 output data</b>                                                                                     |              |
| 0x0A   | GPIODDO        | <b>GPIO PD0~PD5 output data</b>              |      |                                                                                                                | 0x00         |
|        |                | 7-6                                          | NA   | <b>Reserved</b>                                                                                                |              |
|        |                | 5-0                                          | R/W  | <b>PD0~PD5 output data</b>                                                                                     |              |
| 0x0C   | GPIOAIE        | <b>GPIO PA0~PA7 input enable</b>             |      |                                                                                                                | 0x30         |
|        |                | 7-0                                          | R/W  | <b>PA0~PA7 input enable</b>                                                                                    |              |
| 0x0D   | GPIOBIE        | <b>GPIO PB0~PB7 input enable</b>             |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | <b>PB0~PB7 input enable</b>                                                                                    |              |
| 0x0E   | GPIOCIE        | <b>GPIO PC0~PC7 input enable</b>             |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | <b>PC0~PC7 input enable</b>                                                                                    |              |
| 0x0F   | GPIODIE        | <b>GPIO PD0~PD5 input enable</b>             |      |                                                                                                                | 0x00         |
|        |                | 7-6                                          | NA   | <b>Reserved</b>                                                                                                |              |
|        |                | 5-0                                          | R/W  | <b>PD0~PD5 input enable</b>                                                                                    |              |
| 0x11   | GPIOADI        | <b>GPIO PA0~PA7 input data</b>               |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | RO   | <b>PA0~PA7 input data</b>                                                                                      |              |
| 0x12   | GPIOBDI        | <b>GPIO PB0~PB7 input data</b>               |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | RO   | <b>PB0~PB7 input data</b>                                                                                      |              |
| 0x13   | GPIOCDI        | <b>GPIO PC0~PC7 input data</b>               |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | RO   | <b>PC0~PC7 input data</b>                                                                                      |              |
| 0x14   | GPIODDI        | <b>GPIO PD0~PD5 input data</b>               |      |                                                                                                                | 0x00         |
|        |                | 7-6                                          | NA   | <b>Reserved</b>                                                                                                |              |
|        |                | 5-0                                          | RO   | <b>PD0~PD5 input data</b>                                                                                      |              |
| 0x80   | GPIOAE12       | <b>GPIO PA0~PA7 driving strength control</b> |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | Driving control<br>0: weak<br>1: strong                                                                        |              |
| 0x81   | GPIOBE12       | <b>GPIO PB0~PB7 driving strength control</b> |      |                                                                                                                | 0x00         |
|        |                | 7-0                                          | R/W  | Driving control<br>0: weak<br>1: strong                                                                        |              |

| offset | register abbr. | register name                                |      |                                         | pwr-on value |
|--------|----------------|----------------------------------------------|------|-----------------------------------------|--------------|
|        |                | bit                                          | attr | description                             |              |
| 0x82   | GPIOCE12       | <b>GPIO PC0~PC7 driving strength control</b> |      |                                         | 0x00         |
|        |                | 1-0                                          | R/W  | Driving control<br>0: weak<br>1: strong |              |
| 0x83   | GPIODE12       | <b>GPIO PD0~PD5 driving strength control</b> |      |                                         | 0x00         |
|        |                | 7-6                                          | NA   | <b>Reserved</b>                         |              |
|        |                | 5-0                                          | R/W  | Driving control<br>0: weak<br>1: strong |              |
| 0xB0   | PAPU           | <b>PA0~A7 Pull-up control</b>                |      |                                         | 0x3F         |
|        |                | 7-0                                          | R/W  | PA0~PA7 Pull-up Enable                  |              |
| 0xB1   | PBPU           | <b>PB0~B7 Pull-up control</b>                |      |                                         | 0x00         |
|        |                | 7-0                                          | R/W  | PB0~PB7 Pull-up Enable                  |              |
| 0xB2   | PCPU           | <b>PC0~C7 Pull-up control</b>                |      |                                         | 0x00         |
|        |                | 7-0                                          | R/W  | PC0~PC7 Pull-up Enable                  |              |
| 0xB3   | PDPU           | <b>PD0~D5 Pull-up control</b>                |      |                                         | 0x30         |
|        |                | 7-6                                          | NA   | <b>Reserved</b>                         |              |
|        |                | 5-0                                          | R/W  | PD0~PD5 Pull-up Enable                  |              |

## 2.4 GPWU

base address : 0xF200

| offset | register abbr. | register name                         |      |                                                                                                                                       | pwr-on value |
|--------|----------------|---------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                   | attr | description                                                                                                                           |              |
| 0x90   | GPWA0<br>GPWA1 | <b>GPWA0 control of PA0 &amp; PA1</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWA1 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWA1 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWA1 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWA0 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWA0 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWA0 enable                                                                                                                          |              |
| 0x91   | GPWA2<br>GPWA3 | <b>GPWA2 control of PA2 &amp; PA3</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWA3 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWA3 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWA3 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWA2 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWA2 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWA2 enable                                                                                                                          |              |
| 0x92   | GPWA4<br>GPWA5 | <b>GPWA4 control of PA4 &amp; PA5</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWA5 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWA5 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWA5 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWA4 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWA4 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWA4 enable                                                                                                                          |              |

| offset | register abbr. | register name              |      |                                                                                                                                       | pwr-on value |
|--------|----------------|----------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                        | attr | description                                                                                                                           |              |
| 0x93   | GPWA6<br>GPWA7 | GPWA6 control of PA6 & PA7 |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                        | R/W  | GPWA7 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                          | R/W  | GPWA7 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                          | R/W  | GPWA7 enable                                                                                                                          |              |
|        |                | 3-2                        | R/W  | GPWA6 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                          | R/W  | GPWA6 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                          | R/W  | GPWA6 enable                                                                                                                          |              |
| 0x94   | GPWB0<br>GPWB1 | GPWB0 control of PB0 & PB1 |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                        | R/W  | GPWB1 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                          | R/W  | GPWB1 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                          | R/W  | GPWB1 enable                                                                                                                          |              |
|        |                | 3-2                        | R/W  | GPWB0 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                          | R/W  | GPWB0 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                          | R/W  | GPWB0 enable                                                                                                                          |              |
| 0x95   | GPWB2<br>GPWB3 | GPWB2 control of PB2 & PB3 |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                        | R/W  | GPWB3 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                          | R/W  | GPWB3 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                          | R/W  | GPWB3 enable                                                                                                                          |              |
|        |                | 3-2                        | R/W  | GPWB2 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                          | R/W  | GPWB2 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                          | R/W  | GPWB2 enable                                                                                                                          |              |

| offset | register abbr. | register name                         |      |                                                                                                                                       | pwr-on value |
|--------|----------------|---------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                   | attr | description                                                                                                                           |              |
| 0x96   | GPWB4<br>GPWB5 | <b>GPWB4 control of PB4 &amp; PB5</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWB5 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWB5 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWB5 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWB4 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWB4 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWB4 enable                                                                                                                          |              |
| 0x97   | GPWB6<br>GPWB7 | <b>GPWB6 control of PB6 &amp; PB7</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWB7 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWB7 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWB7 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWB6 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWB6 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWB6 enable                                                                                                                          |              |
| 0x98   | GPWC0<br>GPWC1 | <b>GPWC0 control of PC0 &amp; PC1</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWC1 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWC1 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWC1 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWC0 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWC0 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWC0 enable                                                                                                                          |              |

| offset | register abbr. | register name                         |      |                                                                                                                                       | pwr-on value |
|--------|----------------|---------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                   | attr | description                                                                                                                           |              |
| 0x98   | GPWC2<br>GPWC3 | <b>GPWC2 control of PC2 &amp; PC3</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWC3 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWC3 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWC3 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWC2 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWC2 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWC2 enable                                                                                                                          |              |
| 0x9A   | GPWC4<br>GPWC5 | <b>GPWC4 control of PC4 &amp; PC5</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWC5 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWC5 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWC5 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWC4 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWC4 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWC4 enable                                                                                                                          |              |
| 0x9B   | GPWC6<br>GPWC7 | <b>GPWC6 control of PC6 &amp; PC7</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                                   | R/W  | GPWC7 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                                     | R/W  | GPWC7 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                                     | R/W  | GPWC7 enable                                                                                                                          |              |
|        |                | 3-2                                   | R/W  | GPWC6 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                                     | R/W  | GPWC6 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                                     | R/W  | GPWC6 enable                                                                                                                          |              |

| offset | register abbr. | register name              |      |                                                                                                                                       | pwr-on value |
|--------|----------------|----------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                        | attr | description                                                                                                                           |              |
| 0x9C   | GPWD0<br>GPWD1 | GPWD0 control of PD0 & PD1 |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                        | R/W  | GPWD1 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                          | R/W  | GPWD1 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                          | R/W  | GPWD1 enable                                                                                                                          |              |
|        |                | 3-2                        | R/W  | GPWD0 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                          | R/W  | GPWD0 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                          | R/W  | GPWD0 enable                                                                                                                          |              |
| 0x9D   | GPWD2<br>GPWD3 | GPWD2 control of PD2 & PD3 |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                        | R/W  | GPWD3 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                          | R/W  | GPWD3 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                          | R/W  | GPWD3 enable                                                                                                                          |              |
|        |                | 3-2                        | R/W  | GPWD2 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                          | R/W  | GPWD2 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                          | R/W  | GPWD2 enable                                                                                                                          |              |
| 0x9E   | GPWD4<br>GPWD5 | GPWD4 control of PD4 & PD5 |      |                                                                                                                                       | 0x00         |
|        |                | 7-6                        | R/W  | GPWD5 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 5                          | R/W  | GPWD5 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 4                          | R/W  | GPWD5 enable                                                                                                                          |              |
|        |                | 3-2                        | R/W  | GPWD4 polarity select<br>0/1/2/3:<br>positive/negative/both/res. if edge trigger<br>high-active/low-active/res./res. if level trigger |              |
|        |                | 1                          | R/W  | GPWD4 trigger type<br>0/1: edge/level trigger                                                                                         |              |
|        |                | 0                          | R/W  | GPWD4 enable                                                                                                                          |              |

| offset | register abbr. | register name                           |       |                      | pwr-on value |
|--------|----------------|-----------------------------------------|-------|----------------------|--------------|
|        |                | bit                                     | attr  | description          |              |
| 0xA0   | GPWUPFA        | <b>GPWUPA0~PA7 pending flag PA0~PA7</b> |       |                      | 0x00         |
|        |                | 7-0                                     | R/W1C | PA0~PA7 pending flag |              |
| 0xA1   | GPWUPFB        | <b>GPWUPB0~PB7 pending flag PB0~PB7</b> |       |                      | 0x00         |
|        |                | 7-0                                     | R/W1C | PB0~PB7 pending flag |              |
| 0xA2   | GPWUPFC        | <b>GPWUPC0~PC7 pending flag PC0~PC7</b> |       |                      | 0x00         |
|        |                | 7-0                                     | R/W1C | PC0~PC7 pending flag |              |
| 0xA3   | GPWUPFD        | <b>GPWUPD0~PD5 pending flag PD0~PD5</b> |       |                      | 0x00         |
|        |                | 7-6                                     | NA    | Reserved             |              |
|        |                | 5-0                                     | R/W1C | PD0~PD5 pending flag |              |

| offset | register abbr. | register name                               |      |                          | pwr-on value |
|--------|----------------|---------------------------------------------|------|--------------------------|--------------|
|        |                | bit                                         | attr | description              |              |
| 0xA4   | GPWUINTEN A    | <b>GPWUPA0~PA7 Interrupt Enable PA0~PA7</b> |      |                          | 0x00         |
|        |                | 7-0                                         | R/W  | PA0~PA7 Interrupt Enable |              |
| 0xA5   | GPWUINTEN B    | <b>GPWUPB0~PB7 Interrupt Enable PB0~PB7</b> |      |                          | 0x00         |
|        |                | 7-0                                         | R/W  | PB0~PB7 Interrupt Enable |              |
| 0xA6   | GPWUINTEN C    | <b>GPWUPC0~PC7 Interrupt Enable PC0~PC7</b> |      |                          | 0x00         |
|        |                | 7-0                                         | R/W  | PC0~PC7 Interrupt Enable |              |
| 0xA7   | GPWUINTEN D    | <b>GPWUPD0~PD5 Interrupt Enable PD0~PD5</b> |      |                          | 0x00         |
|        |                | 7-6                                         | NA   | Reserved                 |              |
|        |                | 5-0                                         | R/W  | PD0~PD5 Interrupt Enable |              |

## 2.5 PWM

base address : 0xF200

| offset | register abbr. | register name                                     |      |                                       | pwr-on value |
|--------|----------------|---------------------------------------------------|------|---------------------------------------|--------------|
|        |                | bit                                               | attr | description                           |              |
| 0x00   | GPIOMD0        | <b>GPIO PA6, PA7, PB0~PB5 mode select</b>         |      |                                       | 0x00         |
|        |                | 7                                                 | R/W  | PB5 is GPIO or PWM7<br>0/1: GPIO/PWM  |              |
|        |                | 6                                                 | R/W  | PB4 is GPIO or PWM6<br>0/1: GPIO/PWM  |              |
|        |                | 5                                                 | R/W  | PB3 is GPIO or PWM5<br>0/1: GPIO/PWM  |              |
|        |                | 4                                                 | R/W  | PB2 is GPIO or PWM4<br>0/1: GPIO/PWM  |              |
|        |                | 3                                                 | R/W  | PB1 is GPIO or PWM3<br>0/1: GPIO/PWM  |              |
|        |                | 2                                                 | R/W  | PB0 is GPIO or PWM2<br>0/1: GPIO/PWM  |              |
|        |                | 1                                                 | R/W  | PA7 is GPIO or PWM1<br>0/1: GPIO/PWM  |              |
|        |                | 0                                                 | R/W  | PA6 is GPIO or PWM0<br>0/1: GPIO/PWM  |              |
| 0x01   | GPIOMD1        | <b>GPIO PB6~PB7, PC0~PC5 mode select</b>          |      |                                       | 0x00         |
|        |                | 7                                                 | R/W  | PC5 is GPIO or PWM15<br>0/1: GPIO/PWM |              |
|        |                | 6                                                 | R/W  | PC4 is GPIO or PWM14<br>0/1: GPIO/PWM |              |
|        |                | 5                                                 | R/W  | PC3 is GPIO or PWM13<br>0/1: GPIO/PWM |              |
|        |                | 4                                                 | R/W  | PC2 is GPIO or PWM12<br>0/1: GPIO/PWM |              |
|        |                | 3                                                 | R/W  | PC1 is GPIO or PWM11<br>0/1: GPIO/PWM |              |
|        |                | 2                                                 | R/W  | PC0 is GPIO or PWM10<br>0/1: GPIO/PWM |              |
|        |                | 1                                                 | R/W  | PB7 is GPIO or PWM9<br>0/1: GPIO/PWM  |              |
|        |                | 0                                                 | R/W  | PB6 is GPIO or PWM8<br>0/1: GPIO/PWM  |              |
| 0x02   | GPIOMD2        | <b>GPIO PC6~PC7, PD0~PD5, PA0~PA1 mode select</b> |      |                                       | 0x00         |
|        |                | 7                                                 | R/W  | PA1 is GPIO or PWM23<br>0/1: GPIO/PWM |              |
|        |                | 6                                                 | R/W  | PA0 is GPIO or PWM22<br>0/1: GPIO/PWM |              |
|        |                | 5                                                 | R/W  | PD5 is GPIO or PWM21<br>0/1: GPIO/PWM |              |
|        |                | 4                                                 | R/W  | PD4 is GPIO or PWM20<br>0/1: GPIO/PWM |              |
|        |                | 3                                                 | R/W  | PD3 is GPIO or PWM19<br>0/1: GPIO/PWM |              |
|        |                | 2                                                 | R/W  | PD2 is GPIO or PWM18<br>0/1: GPIO/PWM |              |
|        |                | 1                                                 | R/W  | PC7 is GPIO or PWM17<br>0/1: GPIO/PWM |              |
|        |                | 0                                                 | R/W  | PC6 is GPIO or PWM16<br>0/1: GPIO/PWM |              |

| offset | register abbr. | register name                   |      |                       | pwr-on value |
|--------|----------------|---------------------------------|------|-----------------------|--------------|
|        |                | bit                             | attr | description           |              |
| 0x1C   | PWMACLK        | <b>PWM0~PWM5 clock select</b>   |      |                       | 0x00         |
|        |                | 7-0                             | R/W  | 0~255: PWM0_CLK/1~256 |              |
| 0x1D   | PWMBCLK        | <b>PWM6~PWM11 clock select</b>  |      |                       | 0x00         |
|        |                | 7-0                             | R/W  | 0~255: PWM0_CLK/1~256 |              |
| 0x1E   | PWMCCCLK       | <b>PWM12~PWM17 clock select</b> |      |                       | 0x00         |
|        |                | 7-0                             | R/W  | 0~255: PWM0_CLK/1~256 |              |
| 0x1F   | PWMDCLK        | <b>PWM18~PWM23 clock select</b> |      |                       | 0x00         |
|        |                | 7-0                             | R/W  | 0~255: PWM0_CLK/1~256 |              |

| offset | register abbr. | register name                              |      |                                                                                                   | pwr-on value |
|--------|----------------|--------------------------------------------|------|---------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                        | attr | description                                                                                       |              |
| 0x84   | PWMSW          | <b>PWM 0~23 Switch control</b>             |      |                                                                                                   | 0x00         |
|        |                | 7                                          | R/W  | bypass PWM clock gators                                                                           |              |
|        |                |                                            |      | 0/1: gated when all PWM disabled/always runs                                                      |              |
|        |                | 6                                          | R/W  | bypass GPW clock gator                                                                            |              |
| 0x85   | PWMOD1         | <b>PWM 0~7 open-drain output control</b>   |      |                                                                                                   | 0x00         |
|        |                | 7-0                                        | R/W  | for all of these open-drain control registers,<br>0/1: disable/enable<br>(push-pull / open-drain) |              |
| 0x86   | PWMOD2         | <b>PWM 8~15 open-drain output control</b>  |      |                                                                                                   | 0x00         |
|        |                | 7-0                                        | R/W  | for all of these open-drain control registers,<br>0/1: disable/enable<br>(push-pull / open-drain) |              |
| 0x87   | PWMOD3         | <b>PWM 16~23 open-drain output control</b> |      |                                                                                                   | 0x00         |
|        |                | 7-0                                        | R/W  | for all of these open-drain control registers,<br>0/1: disable/enable<br>(push-pull / open-drain) |              |

| offset | register abbr. | register name                      |      |                                                                                                                                                                                                                                                                                                                                                    | pwr-on value |
|--------|----------------|------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                | attr | description                                                                                                                                                                                                                                                                                                                                        |              |
| 0x20   | PWMLN0H        | <b>PWM 0 Hi-byte cycle length</b>  |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-2                                | NA   | Reserved                                                                                                                                                                                                                                                                                                                                           |              |
|        |                | 1-0                                | R/W  | 0/1023~256: 1023/256~1023                                                                                                                                                                                                                                                                                                                          |              |
| 0x21   | PWMLN0L        | <b>PWM 0 Low-byte cycle length</b> |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-0                                | R/W  | PWM0 cycle length setting,<br>0/1~255: 1023/1~255 PWM clock                                                                                                                                                                                                                                                                                        |              |
| 0x22   | PWMDT0H        | <b>PWM 0 Hi-byte duty cycle</b>    |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-2                                | NA   | Reserved                                                                                                                                                                                                                                                                                                                                           |              |
|        |                | 1-0                                | R/W  | 1023~256: 1023/256~1023                                                                                                                                                                                                                                                                                                                            |              |
| 0x23   | PWMDT0L        | <b>PWM 0 Low-byte duty cycle</b>   |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-0                                | R/W  | PWM0 duty cycle setting,<br>0~255: 0~255 PWM clock<br>Note:<br>for all of these PWM cycle length and duty cycle control registers<br>1. a 256-T PWM doesn't support 100% duty cycle<br>2. when duty cycle greater or equals cycle length, 100% duty cycle<br>3. PWM output initial value is defined in GPIO DO, for a duty high PWM, set DO to '1' |              |
| 0x24   | PWMLN1H        | <b>PWM 1 Hi-byte cycle length</b>  |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-2                                | NA   | Reserved                                                                                                                                                                                                                                                                                                                                           |              |
|        |                | 1-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |
| 0x25   | PWMLN1L        | <b>PWM 1 Low-byte cycle length</b> |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |
| 0x26   | PWMDT1H        | <b>PWM 1 Hi-byte duty cycle</b>    |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-2                                | NA   | Reserved                                                                                                                                                                                                                                                                                                                                           |              |
|        |                | 1-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |
| 0x27   | PWMDT1L        | <b>PWM 1 Low-byte duty cycle</b>   |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |
| 0x28   | PWMLN2H        | <b>PWM 2 Hi-byte cycle length</b>  |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-2                                | NA   | Reserved                                                                                                                                                                                                                                                                                                                                           |              |
|        |                | 1-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |
| 0x29   | PWMLN2L        | <b>PWM 2 Low-byte cycle length</b> |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |
| 0x2A   | PWMDT2H        | <b>PWM 2 Hi-byte duty cycle</b>    |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-2                                | NA   | Reserved                                                                                                                                                                                                                                                                                                                                           |              |
|        |                | 1-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |
| 0x2B   | PWMDT2L        | <b>PWM 2 Low-byte duty cycle</b>   |      |                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7-0                                | R/W  |                                                                                                                                                                                                                                                                                                                                                    |              |

| offset | register abbr. | register name                      |      |             | pwr-on value |
|--------|----------------|------------------------------------|------|-------------|--------------|
|        |                | bit                                | attr | description |              |
| 0x2C   | PWMLN3H        | <b>PWM 3 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x2D   | PWMLN3L        | <b>PWM 3 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |
| 0x2E   | PWMDT3H        | <b>PWM 3 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x2F   | PWMDT3L        | <b>PWM 3 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |
| 0x30   | PWMLN4H        | <b>PWM 4 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x31   | PWMLN4L        | <b>PWM 4 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |
| 0x32   | PWMDT4H        | <b>PWM 4 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x33   | PWMDT4L        | <b>PWM 4 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |
| 0x34   | PWMLN5H        | <b>PWM 5 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x35   | PWMLN5L        | <b>PWM 5 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |
| 0x36   | PWMDT5H        | <b>PWM 5 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x37   | PWMDT5L        | <b>PWM 5 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |
| 0x38   | PWMLN6H        | <b>PWM 6 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x39   | PWMLN6L        | <b>PWM 6 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |
| 0x3A   | PWMDT6H        | <b>PWM 6 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                | NA   | Reserved    |              |
|        |                | 1-0                                | R/W  |             |              |
| 0x3B   | PWMDT6L        | <b>PWM 6 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                | R/W  |             |              |

| offset | register abbr. | register name                       |      |             | pwr-on value |
|--------|----------------|-------------------------------------|------|-------------|--------------|
|        |                | bit                                 | attr | description |              |
| 0x3C   | PWMLN7H        | <b>PWM 7 Hi-byte cycle length</b>   |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x3D   | PWMLN7L        | <b>PWM 7 Low-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x3E   | PWMDT7H        | <b>PWM 7 Hi-byte duty cycle</b>     |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x3F   | PWMDT7L        | <b>PWM 7 Low-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x40   | PWMLN8H        | <b>PWM 8 Hi-byte cycle length</b>   |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x41   | PWMLN8L        | <b>PWM 8 Low-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x42   | PWMDT8H        | <b>PWM 8 Hi-byte duty cycle</b>     |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x43   | PWMDT8L        | <b>PWM 8 Low-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x44   | PWMLN9H        | <b>PWM 9 Hi-byte cycle length</b>   |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x45   | PWMLN9L        | <b>PWM 9 Low-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x46   | PWMDT9H        | <b>PWM 9 Hi-byte duty cycle</b>     |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x47   | PWMDT9L        | <b>PWM 9 Low-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x48   | PWMLN10H       | <b>PWM 10 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x49   | PWMLN10L       | <b>PWM 10 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x4A   | PWMDT10H       | <b>PWM 10 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x4B   | PWMDT10L       | <b>PWM 10 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |

| offset | register abbr. | register name                       |      |             | pwr-on value |
|--------|----------------|-------------------------------------|------|-------------|--------------|
|        |                | bit                                 | attr | description |              |
| 0x4C   | PWMLN11H       | <b>PWM 11 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x4D   | PWMLN11L       | <b>PWM 11 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x4E   | PWMDT11H       | <b>PWM 11 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x4F   | PWMDT11L       | <b>PWM 11 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x50   | PWMLN12H       | <b>PWM 12 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x51   | PWMLN12L       | <b>PWM 12 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x52   | PWMDT12H       | <b>PWM 12 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x53   | PWMDT12L       | <b>PWM 12 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x54   | PWMLN13H       | <b>PWM 13 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x55   | PWMLN13L       | <b>PWM 13 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x56   | PWMDT13H       | <b>PWM 13 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x57   | PWMDT13L       | <b>PWM 13 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x58   | PWMLN14H       | <b>PWM 14 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x59   | PWMLN14L       | <b>PWM 14 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x5A   | PWMDT14H       | <b>PWM 14 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x5B   | PWMDT14L       | <b>PWM 14 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |

| offset | register abbr. | register name                       |      |             | pwr-on value |
|--------|----------------|-------------------------------------|------|-------------|--------------|
|        |                | bit                                 | attr | description |              |
| 0x5C   | PWMLN15H       | <b>PWM 15 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x5D   | PWMLN15L       | <b>PWM 15 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x5E   | PWMDT15H       | <b>PWM 15 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x5F   | PWMDT15L       | <b>PWM 15 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x60   | PWMLN16H       | <b>PWM 16 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x61   | PWMLN16L       | <b>PWM 16 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x62   | PWMDT16H       | <b>PWM 16 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x63   | PWMDT16L       | <b>PWM 16 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x64   | PWMLN17H       | <b>PWM 17 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x65   | PWMLN17L       | <b>PWM 17 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x66   | PWMDT17H       | <b>PWM 17 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x67   | PWMDT17L       | <b>PWM 17 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x68   | PWMLN18H       | <b>PWM 18 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x69   | PWMLN18L       | <b>PWM 18 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x6A   | PWMDT18H       | <b>PWM 18 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x6B   | PWMDT18L       | <b>PWM 18 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |

| offset | register abbr. | register name                       |      |             | pwr-on value |
|--------|----------------|-------------------------------------|------|-------------|--------------|
|        |                | bit                                 | attr | description |              |
| 0x6C   | PWMLN19H       | <b>PWM 19 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x6D   | PWMLN19L       | <b>PWM 19 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x6E   | PWMDT19H       | <b>PWM 19 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x6F   | PWMDT19L       | <b>PWM 19 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x70   | PWMLN20H       | <b>PWM 20 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x71   | PWMLN20L       | <b>PWM 20 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x72   | PWMDT20H       | <b>PWM 20 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x73   | PWMDT20L       | <b>PWM 20 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x74   | PWMLN21H       | <b>PWM 21 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x75   | PWMLN21L       | <b>PWM 21 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x76   | PWMDT21H       | <b>PWM 21 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x77   | PWMDT21L       | <b>PWM 21 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x78   | PWMLN22H       | <b>PWM 22 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x79   | PWMLN22L       | <b>PWM 22 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x7A   | PWMDT22H       | <b>PWM 22 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x7B   | PWMDT22L       | <b>PWM 22 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |

| offset | register abbr. | register name                       |      |             | pwr-on value |
|--------|----------------|-------------------------------------|------|-------------|--------------|
|        |                | bit                                 | attr | description |              |
| 0x7C   | PWMLN23H       | <b>PWM 23 Hi-byte cycle length</b>  |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x7D   | PWMLN23L       | <b>PWM 23 Low-byte cycle length</b> |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |
| 0x7E   | PWMDT23H       | <b>PWM 23 Hi-byte duty cycle</b>    |      |             | 0x00         |
|        |                | 7-2                                 | NA   | Reserved    |              |
|        |                | 1-0                                 | R/W  |             |              |
| 0x7F   | PWMDT23L       | <b>PWM 23 Low-byte duty cycle</b>   |      |             | 0x00         |
|        |                | 7-0                                 | R/W  |             |              |

## 2.6 FSMB

base address : 0xF600

| offset | register abbr. | register name                                                      |      |                                                                                                                                                                    | pwr-on value |
|--------|----------------|--------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                                                | attr | description                                                                                                                                                        |              |
| 0x10   | FSMBCFG        | <b>Flexible SMBus Master Configuration</b>                         |      |                                                                                                                                                                    | 0x00         |
|        |                | 7                                                                  | WO   | STOP signal<br>FSMBM will issue STOP signal while this bit is written.                                                                                             |              |
|        |                | 6                                                                  | RSV  | Reserved                                                                                                                                                           |              |
|        |                | 5                                                                  | R/W  | Dynamic gating clock enable (for logic clock)<br>Dynamic gating clock is HW auto control.<br>0: Disable 1: Enable                                                  |              |
|        |                | 4                                                                  | WO   | HW Reset<br>Write one reset, Rest All HW circuit except register value                                                                                             |              |
|        |                | 3                                                                  | R/W  | Bus idle time selected<br>0: 64us<br>1: 128us (for SMBus low clock, ex: 7.8KHz)                                                                                    |              |
|        |                | 2                                                                  | R/W  | Host Notify enable<br>0: Disable 1: Enable                                                                                                                         |              |
|        |                | 1                                                                  | R/W  | Timeout function enable<br>0: Disable 1: Enable                                                                                                                    |              |
|        |                | 0                                                                  | R/W  | FSMBM function enable<br>The all logic controller will be reset when disable this function except register value and host notify function.<br>0: Disable 1: Enable |              |
| 0x11   | FSMBFRT        | <b>Flexible SMBus Master Protocol Packet for Flexible Protocol</b> |      |                                                                                                                                                                    | 0x00         |
|        |                | 7~4                                                                | RSV  | Reserved                                                                                                                                                           |              |
|        |                | 3                                                                  | R/W  | CMD byte<br>0: No CMD byte on protocol<br>1: Has CMD byte on protocol<br>Please prepare this control bit before issuing Flexible protocol.                         |              |
|        |                | 2                                                                  | R/W  | CNT byte<br>0: No CNT byte on protocol<br>1: Has CNT byte on protocol<br>Please prepare this control bit before issuing Flexible protocol.                         |              |
|        |                | 1                                                                  | R/W  | PEC byte<br>0: No PEC byte on protocol<br>1: Has PEC byte on protocol<br>Please prepare this control bit before issuing Flexible protocol.                         |              |
| 0x12   | FSMBPEC        | <b>Flexible SMBus Master PEC Value</b>                             |      |                                                                                                                                                                    | 0x00         |
|        |                | 7~0                                                                | RO   | PEC value from master controller                                                                                                                                   |              |
| 0x13   | FSMBOFH        | <b>Flexible SMBus Master Operating Frequency High</b>              |      |                                                                                                                                                                    | 0x09         |
|        |                | 7                                                                  | NA   | Reserved                                                                                                                                                           |              |
| 0x14   | FSMBOFL        | 6~0                                                                | R/W  | High Length of Clock = (FSMBMxOFH [6:0]+1) * 0.5us<br>Operating Frequency = 1MHz ~ 7.8KHz                                                                          | 0x09         |
|        |                | 7                                                                  | NA   | Reserved                                                                                                                                                           |              |
| 0x14   | FSMBOFL        | <b>Flexible SMBus Master Operating Frequency Low</b>               |      |                                                                                                                                                                    | 0x09         |
|        |                | 6~0                                                                | R/W  | Low Length of Clock = (FSMBMxOFL [6:0]+1) * 0.5us<br>Operating Frequency = 1MHz ~ 7.8KHz                                                                           |              |

| offset | register abbr. | register name                                       |       |                                                                                                                                                                                                                                                                                                 | pwr-on value |
|--------|----------------|-----------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                                 | attr  | description                                                                                                                                                                                                                                                                                     |              |
| 0x15   | FSMBIE         | <b>Flexible SMBus Master Interrupt Enable</b>       |       |                                                                                                                                                                                                                                                                                                 | 0x00         |
|        |                | 7~3                                                 | NA    | Reserve                                                                                                                                                                                                                                                                                         |              |
|        |                | 2                                                   | R/W   | Interrupt Enable of block data array                                                                                                                                                                                                                                                            |              |
|        |                | 1                                                   | R/W   | Interrupt Enable of Host Notify command completed                                                                                                                                                                                                                                               |              |
|        |                | 0                                                   | R/W   | Interrupt Enable of completed                                                                                                                                                                                                                                                                   |              |
| 0x16   | FSMBPF         | <b>Flexible SMBus Master Interrupt Pending Flag</b> |       |                                                                                                                                                                                                                                                                                                 | 0x00         |
|        |                | 7~3                                                 | RSV   | Reserve                                                                                                                                                                                                                                                                                         |              |
|        |                | 2                                                   | R/WC1 | Block data array pending flag<br>If RX/TX data are over 32 bytes, User could control the protocol progress via this bit.<br>0: Block Data Array protocol keeps going.<br>1: Block Data Array protocol stops<br>Prepare data finish and clear this bit then protocol to keep on.                 |              |
|        |                | 1                                                   | R/WC1 | Host Notify Command Complete pending flag<br>This bit is the event for pending flag of Host Notify command complete                                                                                                                                                                             |              |
|        |                | 0                                                   | R/WC1 | Complete pending flag<br>This bit is the event for pending flag of bus completed<br>In SMBus: Protocol completed<br>In I2C/PM Bus: Read/Write protocol complete (for protocol level)                                                                                                            |              |
| 0x17   | FSMBPECB       | <b>Flexible SMBus Master PEC from Bus</b>           |       |                                                                                                                                                                                                                                                                                                 | 0x00         |
|        |                | 7-0                                                 | RO    | The PEC value from bus                                                                                                                                                                                                                                                                          |              |
| 0x18   | FSMBPRTC       | <b>Flexible SMBus Master Protocol</b>               |       |                                                                                                                                                                                                                                                                                                 | 0x00         |
|        |                | 7                                                   | R/W   | Protocol transacts with PEC (Packet Error Check)<br>0: Disable      1: Enable.                                                                                                                                                                                                                  |              |
|        |                | 6-0                                                 | R/W   | Command protocol.<br>02h: Quick Write<br>03h: Quick Read<br>04h: Send Byte<br>05h: Receive Byte<br>06h: Write Byte<br>07h: Read Byte<br>08h: Write Word<br>09h: Read Word<br>0Ah: Write Block<br>0Bh: Read Block<br>0Ch: Word Process<br>0Dh: Block Process<br>7Fh: Flexible Protocol for I2C   |              |
| 0x19   | FSMBSTS        | <b>Flexible SMBus Master Status</b>                 |       |                                                                                                                                                                                                                                                                                                 | 0x00         |
|        |                | 7                                                   | RO    | FSMBM status of Master State Machine<br>0: Busy state      1: Idle state                                                                                                                                                                                                                        |              |
|        |                | 6                                                   | RO    | FSMBM status of Host Notify State Machine<br>0: Busy state      1: Idle state                                                                                                                                                                                                                   |              |
|        |                | 5                                                   | RO    | Reserve                                                                                                                                                                                                                                                                                         |              |
|        |                | 4-0                                                 | RO    | Error code.<br>00h: No error<br>10h: Device address no ACK<br>12h: Command no ACK<br>13h: Device data no ACK<br>17h: Lost arbitration<br>18h: SMBus timeout, SCL low timeout 28ms~32ms<br>19h: Unsupported protocol<br>1Ah: SMBus busy<br>1Eh: STOP fail<br>1Fh: PEC (Packet Error Check) error |              |

| offset            | register abbr. | register name                                      |      |                                                                                                  | pwr-on value |
|-------------------|----------------|----------------------------------------------------|------|--------------------------------------------------------------------------------------------------|--------------|
|                   |                | bit                                                | attr | description                                                                                      |              |
| 0x1A              | FSMBADR        | <b>Flexible SMBus Master Address</b>               |      |                                                                                                  | 0x00         |
|                   |                | 7-1                                                | R/W  | SMBus address (7-bits long).                                                                     |              |
|                   |                | 0                                                  | R/W  | Data direction bit<br>0: Write    1: Read<br>In SMBus protocol (02~0D command) ignored this bit. |              |
| 0x1B              | FSMBCMD        | <b>Flexible SMBus Master Command Port</b>          |      |                                                                                                  | 0x00         |
|                   |                | 7-0                                                | R/W  | Command port                                                                                     |              |
| 0x1C<br>~<br>0x3B | FSMBDATx       | <b>Flexible SMBus Master Data Array (32 Bytes)</b> |      |                                                                                                  | 0x00         |
|                   |                | 7-0                                                | R/W  | Data buffer                                                                                      |              |

## 2.7 I<sup>2</sup>C Slave

base address : 0xF500

| offset | register abbr. | register name                                |      |                                                                                                                                                                                                                                                                                                                                                                    | pwr-on value |
|--------|----------------|----------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                          | attr | description                                                                                                                                                                                                                                                                                                                                                        |              |
| 0x00   | I2CCFG         | <b>I<sup>2</sup>C Slave Configuration</b>    |      |                                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7                                            | R/W  | Pending Flag of the first data in ADR_W function enable (command pending flag)                                                                                                                                                                                                                                                                                     |              |
|        |                | 6                                            | R/W  | Pending Flag of Data Register Double Buffering function enable<br>Note:<br>The Double Buffering function divides the 16-byte data buffer into two 8-byte data buffers which take turns to store incoming or outgoing data. This function improves I <sup>2</sup> C data throughput by eliminating the I <sup>2</sup> C wait-states caused by buffer full condition |              |
|        |                | 5                                            | R/W  | Dynamic gating clock enable (for logic clock)                                                                                                                                                                                                                                                                                                                      |              |
|        |                | 4                                            | R/W  | Timeout2 function enable<br>Note:<br>Timeout2 will occur when SCL is high and SDA sustains low for 28m~32ms.                                                                                                                                                                                                                                                       |              |
|        |                | 3                                            | R/W  | Byte Counter enable<br>0: Disable(default), byte number has no limitation in TX/RX state<br>1:Enable, the maximum of 16bytes for TX/RX data                                                                                                                                                                                                                        |              |
|        |                | 2                                            | R/W  | PEC enable (Packet error checking)<br>The maximum of 16bytes is TX/RX data when PEC enable<br>Note:<br>The byte counter must be enabled when PEC enable, otherwise, PEC byte can't receiver or transmitter in protocol                                                                                                                                             |              |
|        |                | 1                                            | R/W  | Timeout function enable<br>Note:<br>Timeout will occur when SCL signal sustains low for 28m~32ms.                                                                                                                                                                                                                                                                  |              |
|        |                | 0                                            | R/W  | I2C Slave function enable<br>The Slave module and wake up function will be reset when this bit is disable.                                                                                                                                                                                                                                                         |              |
| 0x01   | I2CS0IE        | <b>I<sup>2</sup>C Slave Interrupt Enable</b> |      |                                                                                                                                                                                                                                                                                                                                                                    | 0x00         |
|        |                | 7~6                                          | RSV  | Reserved                                                                                                                                                                                                                                                                                                                                                           |              |
|        |                | 5                                            | R/W  | Interrupt Enable of the first data in ADR_W (command interrupt)<br>This function is working when I2CSxCFG bit7 enabled                                                                                                                                                                                                                                             |              |
|        |                | 4                                            | R/W  | Interrupt Enable of the data register Double Buffering function<br>This function is working when I2CSxCFG bit6 enabled                                                                                                                                                                                                                                             |              |
|        |                | 3                                            | R/W  | Interrupt Enable of Host Notify Completed                                                                                                                                                                                                                                                                                                                          |              |
|        |                | 2                                            | R/W  | Interrupt Enable of Data Register Full                                                                                                                                                                                                                                                                                                                             |              |
|        |                | 1                                            | R/W  | Interrupt Enable of Address ACK                                                                                                                                                                                                                                                                                                                                    |              |
|        |                | 0                                            | R/W  | Interrupt Enable of Protocol Completed                                                                                                                                                                                                                                                                                                                             |              |

| offset | register abbr. | register name                            |       |                                                                                                                                                                                                                                                                                                          | pwr-on value |
|--------|----------------|------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                      | attr  | description                                                                                                                                                                                                                                                                                              |              |
| 0x02   | I2CS0PF        | <b>I<sup>2</sup>C Slave Pending Flag</b> |       |                                                                                                                                                                                                                                                                                                          | 0x00         |
|        |                | 7~6                                      |       | Reserve                                                                                                                                                                                                                                                                                                  |              |
|        |                | 5                                        | R/W1C | Pending Flag of first data in ADDR_W (command pending flag)<br>This bit will be set when the first data is completed.<br>Write one to clear this bit.<br>This function is working when I2CSxCFG bit7 enable<br>In this Pending Flag state, slave will drive SCL to Low                                   |              |
|        |                | 4                                        | R/W1C | Pending Flag of 8-Byte Data Register full in Double Buffering Mode<br>This bit will be set when data register is half(8 bytes)-full<br>Write one to clear this bit.<br>This function is working when I2CSxCFG bit6 enable<br>In this Pending Flag state, slave will drive SCL to Low                     |              |
|        |                | 3                                        | R/W1C | Pending Flag of Host Notify Completed<br>This bit will be set when host notify protocol is completed<br>Write one to clear this bit.                                                                                                                                                                     |              |
|        |                | 2                                        | R/W1C | Pending Flag of Data Register Full<br>This bit will be set when data register is full<br>Write one to clear this bit.<br>In this Pending Flag state, slave will drive SCL to Low                                                                                                                         |              |
|        |                | 1                                        | R/W1C | Pending Flag of Address ACK<br>This bit will be set when address ACK signal is completed<br>Write one to clear this bit.<br>In this Pending Flag state, slave will drive SCL to Low                                                                                                                      |              |
|        |                | 0                                        | R/W1C | Pending Flag of Protocol Completed<br>This bit will be set when protocol is completed.<br>Write one to clear this bit.                                                                                                                                                                                   |              |
| 0x03   | I2CS0STS       | <b>I<sup>2</sup>C Slave Status</b>       |       |                                                                                                                                                                                                                                                                                                          | 0x00         |
|        |                | 7-5                                      | RSV   | Reserved                                                                                                                                                                                                                                                                                                 |              |
|        |                | 4                                        | RO    | SMBus / I2C_Slave controller in idle or busy state<br>0: Idle      1: Busy                                                                                                                                                                                                                               |              |
|        |                | 3~0                                      | RO    | Error Code<br>0h: No Error<br>1h: Timeout when SCL sustains low for 28m~32ms.<br>2h: Timeout2 when SCL is high and SDA sustains low for 28m~32ms.<br>3h: PEC error<br>4h: Protocol Error. The Expected Data is not equal to the Actual Data.                                                             |              |
| 0x04   | I2CS0BA        | <b>I<sup>2</sup>C Slave Base Address</b> |       |                                                                                                                                                                                                                                                                                                          | 0x00         |
|        |                | 7-0                                      | R/W   | Base Address setting<br>Bit0 is ignored                                                                                                                                                                                                                                                                  |              |
| 0x05   | I2CS0MA        | <b>I<sup>2</sup>C Slave Mask Address</b> |       |                                                                                                                                                                                                                                                                                                          | 0x00         |
|        |                | 7-0                                      | R/W   | Address Mask setting (Note: Bit0 is always ignored)<br>Set "1" to mask-off this address bit for slave address matching<br>For example: base address equals 0xAA<br>1. mask address =0x0F, the slave address range = 0xA0~0xAF<br>2. mask address =0x0A, the slave address range = 0xA0, 0xA2, 0xA8, 0xAA |              |

| offset            | register abbr. | register name                                     |      |                                                                                                                                                                                                        | pwr-on value |
|-------------------|----------------|---------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|                   |                | bit                                               | attr | description                                                                                                                                                                                            |              |
| 0x06              | I2CS0CNT       | <b>I<sup>2</sup>C Slave Byte Count</b>            |      |                                                                                                                                                                                                        | 0x00         |
|                   |                | 7~5                                               | RSV  | Reserve                                                                                                                                                                                                |              |
|                   |                | 4~0                                               | R/W  | Programmable TX/RX byte counter number<br>This is valid when byte counter enable                                                                                                                       |              |
| 0x07              | I2CS0CNTB      | <b>I<sup>2</sup>C Slave Byte Count From Bus</b>   |      |                                                                                                                                                                                                        | 0x00         |
|                   |                | 7~5                                               | RSV  | Reserve                                                                                                                                                                                                |              |
|                   |                | 4~0                                               | RO   | Byte counter from bus in TX/RX state<br>This register is additional when data state goes to ACK state.<br>This register is cleared when Address ACK or Data Full pending flag change from high to low. |              |
| 0x08<br>~<br>0x17 | I2CS0DATx      | <b>I<sup>2</sup>C Slave Data Array (16 Bytes)</b> |      |                                                                                                                                                                                                        | 0x00         |
|                   |                | 7-0                                               | R/W  | Data Register for SMB Slave Operation                                                                                                                                                                  |              |
| 0x18              | I2CS0ADRM      | <b>I<sup>2</sup>C Slave Address From Master</b>   |      |                                                                                                                                                                                                        | 0x00         |
|                   |                | 7-1                                               | R/W  | Address from Master                                                                                                                                                                                    |              |
|                   |                | 0                                                 | R/W  | Direction Bit<br><b>0</b> : Write (Master to Slave)<br><b>1</b> : Read (Slave to Master)                                                                                                               |              |

## 2.8 FMC

base address : 0xF800

| offset | register abbr. | register name                           |      |                                                                                                                                                                                                                                                                                                                                                      | pwr-on value |
|--------|----------------|-----------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                     | attr | description                                                                                                                                                                                                                                                                                                                                          |              |
| 0x00   | FMCCFG         | <b>FMC configuration</b>                |      |                                                                                                                                                                                                                                                                                                                                                      | 0xD0         |
|        |                | 7                                       | RO   | XBI controller idle flag<br>0/1: busy/idle                                                                                                                                                                                                                                                                                                           |              |
|        |                | 6-2                                     | NA   | Reserved                                                                                                                                                                                                                                                                                                                                             |              |
|        |                | 1                                       | RO   | XBI register command busy flag<br>0/1: idle/busy                                                                                                                                                                                                                                                                                                     |              |
|        |                | 0                                       | NA   | Reserved                                                                                                                                                                                                                                                                                                                                             |              |
| 0x07   | FMCCMD         | <b>FMC mode command</b>                 |      |                                                                                                                                                                                                                                                                                                                                                      | 0x00         |
|        |                | 7-0                                     | R/W  | write command<br>02h: page latch<br>03h: read<br>20h: erase selected full page (mode = 1011)<br>60h: erase all rows of the bulk (mode = 1001)<br>70h: program selected full page (mode = 1111)<br>80h: clear HVPL data (mode = 0100)<br>90h: read trim data from special rows<br>B0h: program all rows of the bulk (mode = 1100)<br>Others: reserved |              |
| 0x08   | FMCA0          | <b>FMC address [7:0]</b>                |      |                                                                                                                                                                                                                                                                                                                                                      | 0x00         |
|        |                | 7-0                                     | R/W  | address[7:0]                                                                                                                                                                                                                                                                                                                                         |              |
| 0x09   | FMCA1          | <b>FMC address [12:8]</b>               |      |                                                                                                                                                                                                                                                                                                                                                      | 0x00         |
|        |                | 7-5                                     | NA   | Reserved                                                                                                                                                                                                                                                                                                                                             |              |
|        |                | 4-0                                     | R/W  | address [12:8]                                                                                                                                                                                                                                                                                                                                       |              |
| 0x0A   | FMCDO          | <b>FMC Flash data output</b>            |      |                                                                                                                                                                                                                                                                                                                                                      | 0x00         |
|        |                | 7-0                                     | R/W  |                                                                                                                                                                                                                                                                                                                                                      |              |
| 0x0B   | FMCDI          | <b>FMC Flash data input</b>             |      |                                                                                                                                                                                                                                                                                                                                                      | 0x00         |
|        |                | 7-0                                     | RO   |                                                                                                                                                                                                                                                                                                                                                      |              |
| 0x20   | FMCPSA         | <b>FMC Flash protect start address</b>  |      |                                                                                                                                                                                                                                                                                                                                                      | 0xE0         |
|        |                | 7-0                                     | R/W  | Start address = Reg.*128                                                                                                                                                                                                                                                                                                                             |              |
| 0x21   | FMCPEA         | <b>FMC Flash protect end address</b>    |      |                                                                                                                                                                                                                                                                                                                                                      | 0xFF         |
|        |                | 7-0                                     | R/W  | End address = Reg.*128                                                                                                                                                                                                                                                                                                                               |              |
| 0x22   | FMCPAC         | <b>FMC Flash protect access control</b> |      |                                                                                                                                                                                                                                                                                                                                                      | 0x00         |
|        |                | 7                                       | RW   | SMB1 read inhibit                                                                                                                                                                                                                                                                                                                                    |              |
|        |                | 6                                       | R/W  | SMB read inhibit                                                                                                                                                                                                                                                                                                                                     |              |
|        |                | 5                                       | R/W  | EDI read inhibit                                                                                                                                                                                                                                                                                                                                     |              |
|        |                | 4                                       | R/W  | 8051 read inhibit                                                                                                                                                                                                                                                                                                                                    |              |
|        |                | 3                                       | R/W  | SMB1 write inhibit                                                                                                                                                                                                                                                                                                                                   |              |
|        |                | 2                                       | R/W  | SMB write inhibit                                                                                                                                                                                                                                                                                                                                    |              |
|        |                | 1                                       | R/W  | EDI write inhibit                                                                                                                                                                                                                                                                                                                                    |              |
|        |                | 0                                       | R/W  | 8051 write inhibit                                                                                                                                                                                                                                                                                                                                   |              |

- XBI programming note:  
Before writing register 00h/01h/07h~0Ah, be sure (polling) XBIEFCFG [1] is zero.
  
- XBI read flash sequence:  
step1. set axa (0xF801[2]), 0: access main block, 1: access special row  
step2. set address (0xF808/0xF809)  
step3. polling xbi command status (until 0xF800[1]=0)  
step4. fire read command by write 0xF807=0x3  
step5. polling xbi command status (until 0xF800[1]=0) (optional, since read is done very fast)  
step6. read flash data out at 0xF80B

## 2.9 SPI

base address : 0xF900

| offset | register abbr. | register name                                |      |                                                                                                                                                                                                                                                                                                                                                                                        | pwr-on value |
|--------|----------------|----------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                          | attr | description                                                                                                                                                                                                                                                                                                                                                                            |              |
| 0x00   | SHICFG         | <b>SPI host interface configuration</b>      |      |                                                                                                                                                                                                                                                                                                                                                                                        | 0x00         |
|        |                | 7-6                                          | RSV  | Reserved                                                                                                                                                                                                                                                                                                                                                                               |              |
|        |                | 5-4                                          | R/W  | SPI host signal timing relation and SPI mode<br><b>2'b00</b> : Mode 0: Clock default low, Data drive @ falling, latch @ rising<br><b>2'b01</b> : Mode 1: Clock default low, Data drive @ rising, latch @ falling<br><b>2'b10</b> : Mode 2: Clock default high, Data drive @ rising, latch @ falling<br><b>2'b11</b> : Mode 3: Clock default high, Data drive @ falling, latch @ rising |              |
|        |                | 3-1                                          | R/W  | SPI clock selection<br><b>3'b000</b> : SPI clock run 16 MHz<br><b>3'b001</b> : SPI clock run 8 MHz<br><b>3'b010</b> : SPI clock run 4 MHz<br><b>3'b011</b> : SPI clock run 2 MHz<br><b>3'b100</b> : SPI clock run 1MHz<br><b>Other</b> : SPI clock run 0.5 MHz                                                                                                                         |              |
|        |                | 0                                            | R/W  | SPI host controller enable<br><b>0</b> : Disable <b>1</b> : Enable                                                                                                                                                                                                                                                                                                                     |              |
| 0x01   | SHICTR         | <b>SPI host control register</b>             |      |                                                                                                                                                                                                                                                                                                                                                                                        | 0x00         |
|        |                | 7                                            | RO   | SPI host busy flag. If this bit set, the SPI host is in busy state.<br><b>0</b> : Idle <b>1</b> : Busy                                                                                                                                                                                                                                                                                 |              |
|        |                | 6-0                                          | NA   | Reserved                                                                                                                                                                                                                                                                                                                                                                               |              |
| 0x02   | SHITBUF        | <b>SPI host interface transmit data port</b> |      |                                                                                                                                                                                                                                                                                                                                                                                        | 0x00         |
|        |                | 7-0                                          | R/W  | While SHICTR[7]=0 (SPI host is not busy), writing to this register will force data to output on SHIDO in continuously serial 8 bits with MSB first.                                                                                                                                                                                                                                    |              |
| 0x03   | SHIRBUF        | <b>SPI host interface receive data port</b>  |      |                                                                                                                                                                                                                                                                                                                                                                                        | 0x00         |
|        |                | 7-0                                          | RO   | SPI host receive data port.                                                                                                                                                                                                                                                                                                                                                            |              |

## 2.10 SFR

The embedded 8051 is compatible with industrial standard 8051(or 8031). There are 3 standard 8051 peripherals, including the Interrupt controller, the Serial port and two 16-bit timers.

**ENE8051** extends the channels of Interrupt Controller in the original 8051 to 24 channels supporting internal peripheral devices. The Serial port uses **SCON2** to achieve high speed serial transmission rate up to 115,200 bps. The two 16-bit timers are basically the same as that in the standard 8051's, except when **SCON2** is used to generate high-speed baud rate. Under such circumstances the 2 timers will not be used for baud-rate generation and can be used for other purposes.

### Interrupt Vectors Table

| Int. Source | Vector Address | Applications                     | Priority       |
|-------------|----------------|----------------------------------|----------------|
| IE0         | 0x0003         | 8051 external interrupt 0        | 0(Highest)     |
| TF0         | 0x000B         | 8051 Timer 0                     | 1              |
| IE1         | 0x0013         | 8051 external interrupt 1        | 2              |
| TF1         | 0x001B         | 8051 Timer 1                     | 3              |
| RI & TI     | 0x0023         | 8051 Serial port TX/RX interrupt | 4              |
| P0I.[0]     | 0x0043         | Reserved                         | 5              |
| P0I.[1]     | 0x004B         | GPWU PA/PB/PC/PD event interrupt | 6              |
| P0I.[2]     | 0x0053         | WDT timer interrupt              | 7              |
| P0I.[3]     | 0x005B         | LVDA Interrupt                   | 8              |
| P0I.[4]     | 0x0063         | Reserved                         | 9              |
| P0I.[5]     | 0x006B         | Reserved                         | 10             |
| P0I.[6]     | 0x0073         | Reserved                         | 11             |
| P0I.[7]     | 0x007B         | Reserved                         | 12             |
| P1I.[0]     | 0x0083         | Reserved                         | 13             |
| P1I.[1]     | 0x008B         | Reserved                         | 14             |
| P1I.[2]     | 0x0093         | Reserved                         | 15             |
| P1I.[3]     | 0x009B         | Reserved                         | 16             |
| P1I.[4]     | 0x00A3         | FSMB interrupt                   | 17             |
| P1I.[5]     | 0x00AB         | Reserved                         | 18             |
| P1I.[6]     | 0x00B3         | Reserved                         | 19             |
| P1I.[7]     | 0x00BB         | Reserved                         | 20<br>(Lowest) |

\* Interrupt vector = {PCON[5:4], 6'h00, original\_vector\_address[7:0]}

## Special Function Register (SFR)

The Special Function Registers are located in the internal RAM of 8051 microprocessor. The internal address is from 0x80 to 0xFF, sized with 128 bytes. All the SFRs are compatible with the standard ones. Some SFRs are redesigned with new features for flexible application. The following table gives a brief summary.

P1IE, P0IE are read/write registers used as Interrupt Enable (IE) to their corresponding interrupt inputs. These two registers are original 8051 port registers with 8-bits. For the embedded 8051 inside ene6K5830, the 3 ports are used for interrupt input (always rise pulses) extensions. The overall interrupt events are 24.

P1IF, P0IF are Interrupt Flag (IF) corresponding to the 24 interrupt inputs. The IFs are set by external interrupt event (always a rising pulse, one clock width), and are cleared by software (execute IRET instruction for active interrupt).

For more detail, please refer to the section of register description.

### Color information:

|     |                                            |
|-----|--------------------------------------------|
| XXX | Original Industrial standard 8051 features |
| XXX | ENE's embedded 8051 new features           |
| XXX | XXX Changed 8051 feature for ENE 8051      |

|    | 0/8  | 1/9   | 2/A   | 3/B   | 4/C    | 5/D    | 6/E    | 7/F   |    |
|----|------|-------|-------|-------|--------|--------|--------|-------|----|
| F8 |      |       |       |       |        |        |        |       | FF |
| F0 | B    |       |       |       |        |        |        |       | F7 |
| E8 | PE   |       |       |       |        |        |        |       | EF |
| E0 | ACC  |       |       |       |        |        |        |       | E7 |
| D8 | P0IF | P1IF  |       |       |        |        |        |       | DF |
| D0 | PSW  |       |       |       |        |        |        |       | D7 |
| C8 | PD   |       |       |       |        |        |        |       | CF |
| C0 | PC   |       |       |       |        |        |        |       | C7 |
| B8 | IP   |       |       |       |        |        |        |       | BF |
| B0 | PB   |       |       |       |        |        |        |       | B7 |
| A8 | IE   | SPCON |       |       |        |        |        |       | AF |
| A0 | P2   | PCB0  | PCB1  | STS   | IPRTY0 | IPRTY1 | IPRTY2 | ST    | A7 |
| 98 | SCON | SBUF  | SCON2 | SCON3 | SCON4  | INST1  | INST2  | INST3 | 9F |
| 90 | PA   |       |       |       |        |        |        |       | 97 |
| 88 | TCON | TMOD  | TL0   | TL1   | TH0    | TH1    | PCL    | PCH   | 8F |
| 80 |      | SP    | DPL   | DPH   |        |        | PCON2  | PCON  | 87 |
| ★  | 0/8  | 1/9   | 2/A   | 3/B   | 4/C    | 5/D    | 6/E    | 7/F   |    |

➤ The registers listed in the column with ★ mark are all bit addressable.

## SFR Descriptions (Direct Addressing 80h~FFh)

base address : 0xFF00

| offset | register abbr. | register name                       |      |                                                                                                                                                                                                                                                                                              | pwr-on value |
|--------|----------------|-------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                 | attr | description                                                                                                                                                                                                                                                                                  |              |
| 0x80   | P0IE           | <b>Port 0 IE</b>                    |      |                                                                                                                                                                                                                                                                                              | 0x00         |
|        |                | 7-0                                 | R/W  | P0 Interrupt Enable                                                                                                                                                                                                                                                                          |              |
| 0x81   | SP             | <b>Stack Pointer</b>                |      |                                                                                                                                                                                                                                                                                              | 0x00         |
|        |                | 7-0                                 | R/W  | Stack Pointer                                                                                                                                                                                                                                                                                |              |
| 0x82   | DPL            | <b>DPTR Low Byte</b>                |      |                                                                                                                                                                                                                                                                                              | 0x00         |
|        |                | 7-0                                 | R/W  | DPTR low byte                                                                                                                                                                                                                                                                                |              |
| 0x83   | DPH            | <b>DPTR High Byte</b>               |      |                                                                                                                                                                                                                                                                                              | 0x00         |
|        |                | 7-0                                 | R/W  | DPTR high byte                                                                                                                                                                                                                                                                               |              |
| 0x86   | PCON2          | <b>Processor Control Register 2</b> |      |                                                                                                                                                                                                                                                                                              | 0x30         |
|        |                | 7                                   | R/W  | Enable level trigger interrupt                                                                                                                                                                                                                                                               |              |
|        |                | 6                                   | R/W  | TTST, Timer 0/1 test mode, let timer 12 times faster.                                                                                                                                                                                                                                        |              |
|        |                | 5                                   | R/W  | 0: enter stop mode will reset PC<br>1: enter stop mode will not reset PC                                                                                                                                                                                                                     |              |
|        |                | 4                                   | R/W  | Enable external space write.                                                                                                                                                                                                                                                                 |              |
|        |                | 3                                   | R/W  | Next Interrupt Coming Flag.<br>The same extended interrupt coming during ISR before IRET.<br>After exit ISR with IRET instruction, the 8051 will re-enter the ISR again if the flag is 1.<br>Write 0 to clear the flag and prevent from 8051 re-entering the interrupt again after exit ISR. |              |
|        |                | 2                                   | R/W  | PC address selection after wakeup from stop<br>This bit works when PCON2.5 = 1<br>0: 0x0000<br>1: 0x0033                                                                                                                                                                                     |              |
|        |                | 1                                   | R/W  | Enable timer base is 1us                                                                                                                                                                                                                                                                     |              |
|        |                | 0                                   | NA   | Reserved                                                                                                                                                                                                                                                                                     |              |
| 0x87   | PCON           | <b>Processor Control Register</b>   |      |                                                                                                                                                                                                                                                                                              | 0x00         |
|        |                | 7-6                                 | NA   | Reserved                                                                                                                                                                                                                                                                                     |              |
|        |                | 5                                   | R/W  | IVHB, Interrupt vector highest bit. Let interrupt vector to be 00xxh or 80xxh, including standard and extended interrupt.                                                                                                                                                                    |              |
|        |                | 4                                   | R/W  | IVHB2, Interrupt vector 2 <sup>nd</sup> highest bit. Let interrupt vector to be 00xxh or 40xxh, including standard and extended interrupt.                                                                                                                                                   |              |
|        |                | 3-2                                 | NA   | Reserved                                                                                                                                                                                                                                                                                     |              |
|        |                | 1                                   | W1O  | Power Down Mode                                                                                                                                                                                                                                                                              |              |
|        |                | 0                                   | NA   | Reserved                                                                                                                                                                                                                                                                                     |              |

| offset           | register abbr. | register name                  |      |                                                                                                                                                                   | pwr-on value |
|------------------|----------------|--------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|                  |                | bit                            | attr | description                                                                                                                                                       |              |
| 0x88             | TCON           | Timer/Counter Control Register |      |                                                                                                                                                                   | 0x00         |
|                  |                | 7                              | R/WC | TF1<br>Timer 1 overflow flag.                                                                                                                                     |              |
|                  |                | 6                              | R/W  | TR1<br>Timer 1 run control bit                                                                                                                                    |              |
|                  |                | 5                              | R/WC | TF0<br>Timer 0 overflow flag                                                                                                                                      |              |
|                  |                | 4                              | R/W  | TR0<br>Timer 0 run control bit                                                                                                                                    |              |
|                  |                | 3                              | R    | IE1<br>External Interrupt 1 edge detected flag.<br>This flag is clear by HW (when IT1 is high at end of ISR)..                                                    |              |
|                  |                | 2                              | R/W  | IT1<br>Interrupt 1 falling edge / low active control bit                                                                                                          |              |
|                  |                | 1                              | R    | IE0<br>External Interrupt 0 edge detected flag.<br>This flag is clear by HW (when IT0 is high at end of ISR).                                                     |              |
|                  |                | 0                              | R/W  | IT0<br>Interrupt 0 falling edge / low active control bit                                                                                                          |              |
|                  |                | 0x89                           | TMOD | Timer Mode Register                                                                                                                                               |              |
| 7                | R/W            |                                |      | GATE1<br>0: Timer 0 runs when TM1 is 1<br>1: Timer 0 runs when TM1 is 1 and INT1 is high                                                                          |              |
| 6                | R/W            |                                |      | CT1<br>0: be a timer 1. 1: be a counter 1.                                                                                                                        |              |
| 4-5              | R/W            |                                |      | TM1<br>0: Timer 1 is 13-bit timer (8048 timer).<br>1: Timer 1 is 16-bit timer<br>2: Timer 1 is 8-bit auto-reload timer<br>3: Timer 1 is stop.                     |              |
| 3                | R/W            |                                |      | GATE0<br>0: Timer 0 runs when TM0 is 1<br>1: Timer 0 runs when TM0 is 1 and INT0 is high                                                                          |              |
| 2                | R/W            |                                |      | CT0<br>0: be a timer 0. 1: be a counter 0.                                                                                                                        |              |
| 0-1              | R/W            |                                |      | TM0<br>0: Timer 0 is 13-bit timer (8048 timer).<br>1: Timer 0 is 16-bit timer<br>2: Timer 0 is 8-bit auto-reload timer<br>3: Timer 0 TL0, TH0 is two 8-bit timer. |              |
| Timer 0 Low Byte |                |                                |      |                                                                                                                                                                   |              |
| 0x8A             | TL0            | 7-0                            | R/W  | Timer 0 Low Byte                                                                                                                                                  | 0x00         |
| 0x8B             | TL1            | Timer 1 Low Byte               |      |                                                                                                                                                                   | 0x00         |
|                  |                | 7-0                            | R/W  | Timer 1 Low Byte                                                                                                                                                  |              |

| offset       | register abbr. | register name                                 |      |                                                                                                                                                                                    | pwr-on value |
|--------------|----------------|-----------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|              |                | bit                                           | attr | description                                                                                                                                                                        |              |
| 0x8C         | TH0            | <b>Timer 0 high Byte</b>                      |      |                                                                                                                                                                                    | 0x00         |
|              |                | 7-0                                           | R/W  | Timer 0 High Byte                                                                                                                                                                  |              |
| 0x8D         | TH1            | <b>Timer 1 High Byte</b>                      |      |                                                                                                                                                                                    | 0x00         |
|              |                | 7-0                                           | R/W  | Timer 1 High Byte                                                                                                                                                                  |              |
| 0x8E         | PCL            | <b>Program Counter Low Byte</b>               |      |                                                                                                                                                                                    | 0x00         |
|              |                | 7-0                                           | RO   | Program Counter Low Byte                                                                                                                                                           |              |
| 0x8F         | PCH            | <b>Program Counter High Byte</b>              |      |                                                                                                                                                                                    | 0x00         |
|              |                | 7-0                                           | RO   | Program Counter High Byte                                                                                                                                                          |              |
| 0x98         | SCON           | <b>Serial Port Control Register</b>           |      |                                                                                                                                                                                    | 0x40         |
|              |                | 7~6                                           | R/W  | SM[1:0]<br>Serial Port Mode:<br>00: 8-bit shift register mode, E51RX should be E51CLK shift clock.<br>01: 8-bit Serial Port(variable)<br>1X: 9-bit Serial Port (variable)          |              |
|              |                | 5                                             | NA   | Reserved                                                                                                                                                                           |              |
|              |                | 4                                             | R/W  | REN<br>Enable Serial Port reception                                                                                                                                                |              |
|              |                | 3                                             | R/W  | TB8<br>The 9 <sup>th</sup> bit of transmitted in mode 2 & 3.                                                                                                                       |              |
|              |                | 2                                             | R/W  | RB8<br>The 9 <sup>th</sup> bit of received.                                                                                                                                        |              |
|              |                | 1                                             | R/W  | TI<br>Transmit Interrupt flag.                                                                                                                                                     |              |
|              |                | 0                                             | R/W  | RI<br>Receive Interrupt flag.                                                                                                                                                      |              |
| 0x99         | SBUF           | <b>Serial Port Data Buffer</b>                |      |                                                                                                                                                                                    | 0x00         |
|              |                | 7-0                                           | R/W  | Serial port data buffer                                                                                                                                                            |              |
| 0x9A<br>0x9B | SCON2<br>SCON3 | <b>Serial Port Control 2 / 3 Register</b>     |      |                                                                                                                                                                                    | 0x00<br>0x89 |
|              |                | 7-0                                           | R/W  | SCON2 is high byte, SCON3 is low byte to be a 16-bit counter for baud rate based on 8051 clock.<br>Baud rate = (1/T)<br>$T = (8051 \text{ clk period}) * [ \{SCON2, SCON3\} + 1 ]$ |              |
| 0x9C         | SCON4          | <b>Serial Port mode 0 baud- rate Register</b> |      |                                                                                                                                                                                    | 0x00         |
|              |                | 7~2                                           | NA   | Reserved                                                                                                                                                                           |              |
|              |                | 1~0                                           | R/W  | Serial Port mode 0 baud- rate setting<br>0: E51 clock divide 2<br>1: E51 clock divide 4<br>2: E51 clock divide 8<br>3: E51 clock divide 16                                         |              |

| offset | register abbr. | register name                                |      |                                                                                                                                       | pwr-on value |
|--------|----------------|----------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
|        |                | bit                                          | attr | description                                                                                                                           |              |
| 0xA4   | IPRTY0         | <b>Interrupt 2-level Priority Register 0</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7-5                                          | NA   | Reserved                                                                                                                              |              |
|        |                | 4                                            | R/W  | RI & TI interrupt priority<br>0: Low, 1: High                                                                                         |              |
|        |                | 3                                            | R/W  | TF1 interrupt priority                                                                                                                |              |
|        |                | 2                                            | R/W  | IE1 interrupt priority                                                                                                                |              |
|        |                | 1                                            | R/W  | TF0 interrupt priority                                                                                                                |              |
|        |                | 0                                            | R/W  | IE0 interrupt priority                                                                                                                |              |
| 0xA5   | IPRTY1         | <b>Interrupt 2-level Priority Register 1</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7                                            | R/W  | SPI interrupt priority<br>0: Low, 1: High                                                                                             |              |
|        |                | 6-4                                          | NA   | Reserved                                                                                                                              |              |
|        |                | 3                                            | R/W  | LVDA interrupt priority                                                                                                               |              |
|        |                | 2                                            | R/W  | WDT interrupt priority                                                                                                                |              |
|        |                | 1                                            | R/W  | GPWU0/1/2/3 interrupt priority                                                                                                        |              |
| 0xA6   | IPRTY2         | <b>Interrupt 2-level Priority Register 2</b> |      |                                                                                                                                       | 0x00         |
|        |                | 7                                            | R/W  | interrupt priority<br>0: Low, 1: High                                                                                                 |              |
|        |                | 6                                            | R/W  | interrupt priority                                                                                                                    |              |
|        |                | 5                                            | R/W  | interrupt priority                                                                                                                    |              |
|        |                | 4                                            | R/W  | interrupt priority                                                                                                                    |              |
|        |                | 3                                            | R/W  | interrupt priority                                                                                                                    |              |
|        |                | 2                                            | R/W  | interrupt priority                                                                                                                    |              |
|        |                | 1                                            | R/W  | interrupt priority                                                                                                                    |              |
| 0xA8   | IE             | <b>Interrupt Enable Register</b>             |      |                                                                                                                                       | 0x00         |
|        |                | 7                                            | R/W  | EA<br>Disable all interrupt (include extended) if clear to 0. If set to 1, all interrupts should be enabled by individual enable bit. |              |
|        |                | 6-5                                          | NA   | Reserved                                                                                                                              |              |
|        |                | 4                                            | R/W  | ES<br>Enable Serial Port interrupt                                                                                                    |              |
|        |                | 3                                            | R/W  | ET1<br>Enable Timer 1 Overflow interrupt                                                                                              |              |
|        |                | 2                                            | R/W  | EX1<br>Enable External Interrupt 1                                                                                                    |              |
|        |                | 1                                            | R/W  | ET0<br>Enable Timer 0 Overflow interrupt                                                                                              |              |
|        |                | 0                                            | R/W  | EX0<br>Enable External Interrupt 0                                                                                                    |              |

| offset | register abbr. | register name                      |      |                                            | pwr-on value |
|--------|----------------|------------------------------------|------|--------------------------------------------|--------------|
|        |                | bit                                | attr | description                                |              |
| 0xB8   | IP             | <b>Interrupt Priority Register</b> |      |                                            | 0x00         |
|        |                | 7~5                                | NA   | reserved                                   |              |
|        |                | 4                                  | R/W  | PS<br>Serial Port interrupt priority level |              |
|        |                | 3                                  | R/W  | PT1<br>Timer 1 interrupt priority level    |              |
|        |                | 2                                  | R/W  | PX1<br>External Interrupt 1 priority level |              |
|        |                | 1                                  | R/W  | PT0<br>Timer 0 interrupt priority level    |              |
|        |                | 0                                  | R/W  | PX0<br>External Interrupt 0 priority level |              |
| 0xD0   | PSW            | <b>Processor Status Word</b>       |      |                                            | 0x00         |
|        |                | 7                                  | R/W  | CY<br>Carry flag                           |              |
|        |                | 6                                  | R/W  | AC<br>Auxiliary Carry flag                 |              |
|        |                | 5                                  | R/W  | F0<br>Flag 0, for user general purpose.    |              |
|        |                | 4                                  | R/W  | RS1<br>Register Bank selector 1            |              |
|        |                | 3                                  | R/W  | RS0<br>Register Bank selector 0            |              |
|        |                | 2                                  | R/W  | OV<br>Overflow flag                        |              |
|        |                | 1                                  | R/W  | F1<br>Flag 1, for user general purpose.    |              |
| 0xD8   | P0IF           | <b>Interrupt Port 0 Flag</b>       |      |                                            | 0x00         |
|        |                | 7-0                                | R/W  | Interrupt P0 Flag Register                 |              |
| 0xD9   | P1IF           | <b>Interrupt Port 1 Flag</b>       |      |                                            | 0x00         |
|        |                | 7-0                                | R/W  | Interrupt P1 Flag Register                 |              |
| 0xE0   | ACC            | <b>ACC, A Register</b>             |      |                                            | 0x00         |
|        |                | 7-0                                | R/W  | Accumulator                                |              |
| 0xF0   | B              | <b>B Register</b>                  |      |                                            | 0x00         |
|        |                | 7-0                                | R/W  | B Register<br>For MUL and DIV operations.  |              |

## 4. Electronic Characteristics

### 4.1 Absolute Maximum Rating

| Symbol            | Parameter             | Min  | Typ | Max | Unit |
|-------------------|-----------------------|------|-----|-----|------|
| T <sub>A</sub>    | Operating Temperature | -40  | --  | 85  | °C   |
| T <sub>STG</sub>  | Storage Temperature   | -55  | --  | 125 | °C   |
| VDD               | DC Power Voltage      | 2.7  | --- | 5.5 | V    |
| GND               | Power Ground          | -0.3 | 0   | 0.3 | V    |
| IDD               | Max. IDD              | ---  | --- | 200 | mA   |
| I <sub>sink</sub> | Total sink current    | ---  | --- | 320 | mA   |
| ESD               | Human Body Mode       | --   | --  | 8K  | V    |
| EFT               | --                    | --   | --  | 4K  | V    |

### 4.2 DC Electrical Characteristics

| Symbol           | Parameter                                                                  | Condition  | Min | Typ | Max  | Unit |
|------------------|----------------------------------------------------------------------------|------------|-----|-----|------|------|
| VDD              | Operating Voltage                                                          | --         | 2.7 | --  | 5.5V | V    |
| V <sub>POR</sub> | Reset Voltage                                                              | --         | 1.9 | 2.1 | 2.3  | V    |
| I <sub>OP1</sub> | Operating Current @Normal Mode<br>(F <sub>SYS</sub> = 14 MHz, all disable) | VDD = 5.0V | --  | 4.0 | 6.0  | mA   |
|                  |                                                                            | VDD = 3.3V | --  | 3.0 | 4.5  |      |
| I <sub>std</sub> | Standby Current @Sleep Mode(I/O no load, F <sub>SYS</sub> stop)            | VDD = 5.0V | --  | 150 | 180  | μA   |
|                  |                                                                            | VDD = 3.3V | --  | 100 | 150  |      |
| V <sub>IL1</sub> | Input Low Voltage<br>(Except PA0 ~ PA3)                                    | VDD = 5.0V | 0   | --  | 1.5  | V    |
|                  |                                                                            | VDD = 3.3V | 0   | --  | 1.0  |      |
| V <sub>IL2</sub> | Input Low Voltage<br>(PA0 ~ PA3)                                           | VDD = 5.0V | 0   | --  | 0.8  | V    |
|                  |                                                                            | VDD = 3.3V | 0   | --  | 0.8  |      |
| V <sub>IH1</sub> | Input High Voltage<br>(Except PA0 ~ PA3)                                   | VDD = 5.0V | 3.5 | --  | VDD  | V    |
|                  |                                                                            | VDD = 3.3V | 2.3 | --  | VDD  |      |
| V <sub>IH2</sub> | Input High Voltage<br>(PA0 ~ PA3)                                          | VDD = 5.0V | 2.1 | --  | VDD  | V    |
|                  |                                                                            | VDD = 3.3V | 2.1 | --  | VDD  |      |

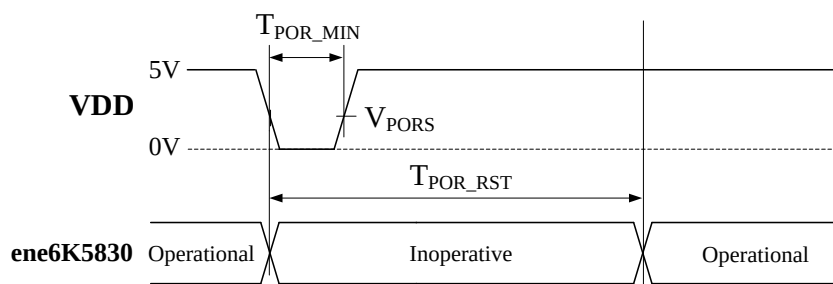
| Symbol    | Parameter                                                    | Condition                | Min | Typ | Max | Unit |
|-----------|--------------------------------------------------------------|--------------------------|-----|-----|-----|------|
| $V_{IH1}$ | Input High Voltage<br>(Except PA0 ~ PA3)                     | VDD = 5.0V               | 3.5 | --  | VDD | V    |
|           |                                                              | VDD = 3.3V               | 2.3 | --  | VDD |      |
| $V_{IH2}$ | Input High Voltage<br>(PA0 ~ PA3)                            | VDD = 5.0V               | 2.1 | --  | VDD | V    |
|           |                                                              | VDD = 3.3V               | 2.1 | --  | VDD |      |
| $I_{OH}$  | I/O Port Source Current                                      | VDD = 5.0V<br>VOH = 4.5V | --  | --  | 10  | mA   |
|           |                                                              | VDD = 3.3V<br>VOH = 2.8V | --  | --  | 5   |      |
| $I_{OL1}$ | I/O Port Sink Current                                        | VDD = 5.0V<br>VOL = 0.5V | --  | --  | 10  | mA   |
|           |                                                              | VDD = 3.3V<br>VOL = 0.3V | --  | --  | 5   |      |
| $I_{OL2}$ | I/O Port Sink Current<br>(Turn on PWM0 ~ PWM8 strong drive)  | VDD = 5.0V<br>VOL = 2.0V | --  | --  | 50  | mA   |
| $I_{OL3}$ | I/O Port Sink Current<br>(Turn on PWM9 ~ PWM23 strong drive) | VDD = 5.0V<br>VOL = 2.0V | --  | --  | 20  | mA   |
| $R_{PH}$  | Pull High resistance                                         | VDD=5.0V                 | --  | 30  | --  | Kohm |
|           |                                                              | VDD=3.3V                 | --  | 50  | --  | Kohm |

#### 4.3 AC Characteristics (TA = -40 ~ 85°C)

| Symbol       | Parameter                 | Condition                            | Min  | Typ  | Max  | Unit |
|--------------|---------------------------|--------------------------------------|------|------|------|------|
| $F_{HOSC}$   | HIRC oscillator frequency | VCC = 4.5 ~ 5.5V                     | 27.2 | 28.0 | 28.8 | MHz  |
| $T_{SYSClk}$ | Oscillator Startup Time   | VCC = 5V<br>$F_{SYS} = 14\text{MHz}$ | --   | --   | 1.5  | μs   |

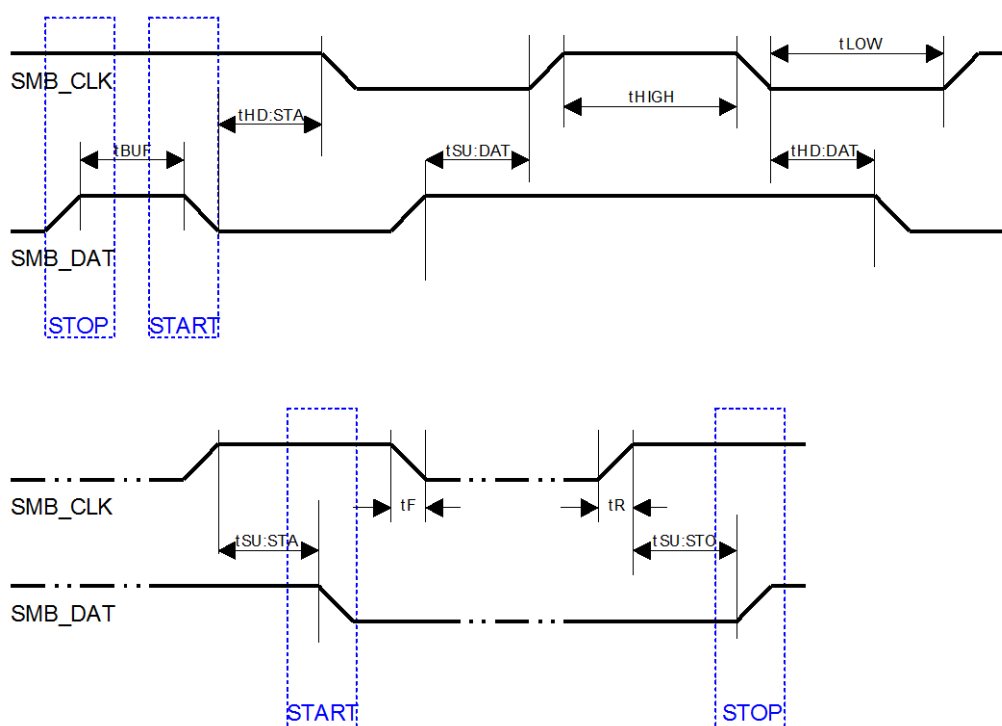
#### 4.4 Power-on Reset Characteristics (TA=25°C, VCC = 5V)

| Symbol         | Parameter                                                 | Condition                | Min | Typ | Max | Unit |
|----------------|-----------------------------------------------------------|--------------------------|-----|-----|-----|------|
| $V_{PORS}$     | Reset Voltage                                             | --                       | 1.9 | 2.1 | 2.3 | V    |
| $T_{POR\_MIN}$ | Power-on Low Level<br>Time after Power-off                | $F_{SYS} = 14\text{MHz}$ | 1.0 | --  | --  | mS   |
| $T_{POR\_RST}$ | Conversion Time for<br>Reeset to Normal<br>Operating Mode | $F_{SYS} = 14\text{MHz}$ | 100 | --  | --  | mS   |



Reset Sequence

## 4.5 SMBus interface Timing



SMBus Timming

### Timing Parameters (SCK, SDA)

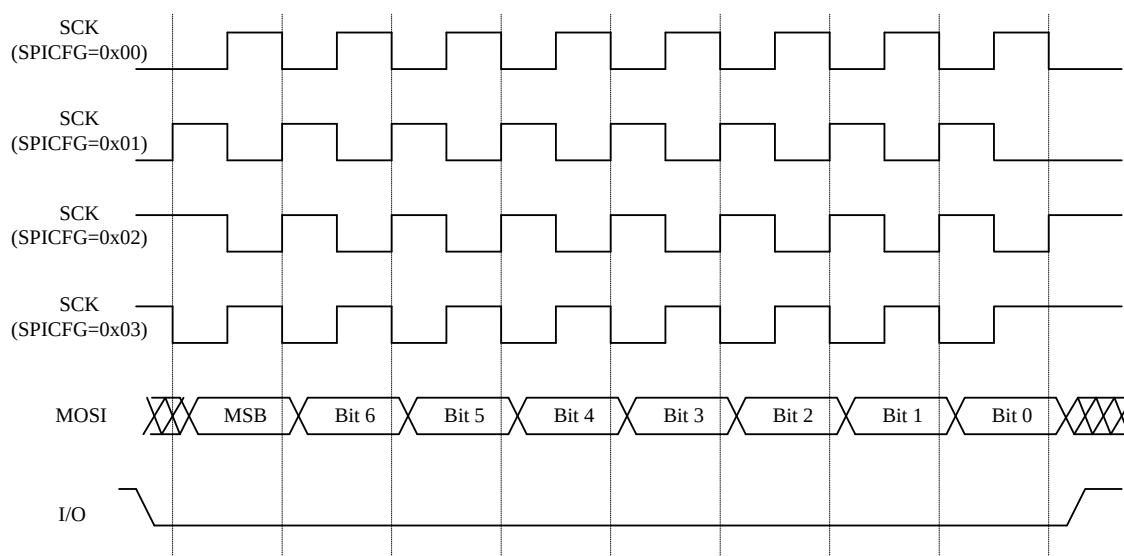
| Symbol       | Parameter                                                                                   | Min | Typ. | Max | Units   | Notes |
|--------------|---------------------------------------------------------------------------------------------|-----|------|-----|---------|-------|
| $T_{buf}$    | Bus free time between Stop and Start Condition                                              | 1.7 | --   | --  | $\mu s$ |       |
| $T_{hd:sta}$ | Hold time after (repeated) star condition. After this period, the first clock is generated. | 1.0 | --   | --  | $\mu s$ |       |
| $T_{su:sta}$ | Repeated start condition setup time                                                         | 1.7 | --   | --  | $\mu s$ |       |
| $T_{su:sto}$ | Stop confition setup time                                                                   | 1.0 | --   | --  | $\mu s$ |       |
| $T_{hd:dat}$ | Data hold time                                                                              | 300 | --   | --  | ns      |       |

| Symbol        | Parameter                | Min | Typ. | Max  | Units   | Notes |
|---------------|--------------------------|-----|------|------|---------|-------|
| $T_{su:dat}$  | Data setup time          | 250 | --   | --   | ns      |       |
| $T_{timeout}$ | Detect clock low timeout | 25  | --   | 35   | ms      |       |
| $T_{low}$     | Clock low period         | 1.3 | --   | --   | $\mu$ s |       |
| $T_{high}$    | Clock high period        | 1.2 | --   | 10   | $\mu$ s |       |
| $T_f$         | Data fall time           | --  | --   | 300  | ns      |       |
| $T_r$         | Data rise time           | --  | --   | 1000 | ns      |       |

#### Timing Parameters (MCK, MDA)

| Symbol        | Parameter                                                                                    | Min | Typ. | Max  | Units   | Notes |
|---------------|----------------------------------------------------------------------------------------------|-----|------|------|---------|-------|
| $T_{buf}$     | Bus free time between Stop and Start Condition                                               | 4.7 | --   | --   | $\mu$ s |       |
| $T_{hd:sta}$  | Hold time after (repeated) start condition. After this period, the first clock is generated. | 4.0 | --   | --   | $\mu$ s |       |
| $T_{su:sta}$  | Repeated start condition setup time                                                          | 4.7 | --   | --   | $\mu$ s |       |
| $T_{su:sto}$  | Stop condition setup time                                                                    | 4.0 | --   | --   | $\mu$ s |       |
| $T_{hd:dat}$  | Data hold time                                                                               | 300 | --   | --   | ns      |       |
| $T_{su:dat}$  | Data setup time                                                                              | 250 | --   | --   | ns      |       |
| $T_{timeout}$ | Detect clock low timeout                                                                     | 25  | --   | 35   | ms      |       |
| $T_{low}$     | Clock low period                                                                             | 4.7 | --   | --   | $\mu$ s |       |
| $T_{high}$    | Clock high period                                                                            | 4.0 | --   | 50   | $\mu$ s |       |
| $T_f$         | Data fall time                                                                               | --  | --   | 300  | ns      |       |
| $T_r$         | Data rise time                                                                               | --  | --   | 1000 | ns      |       |

#### 4.6 SPI Serial Clock Timing



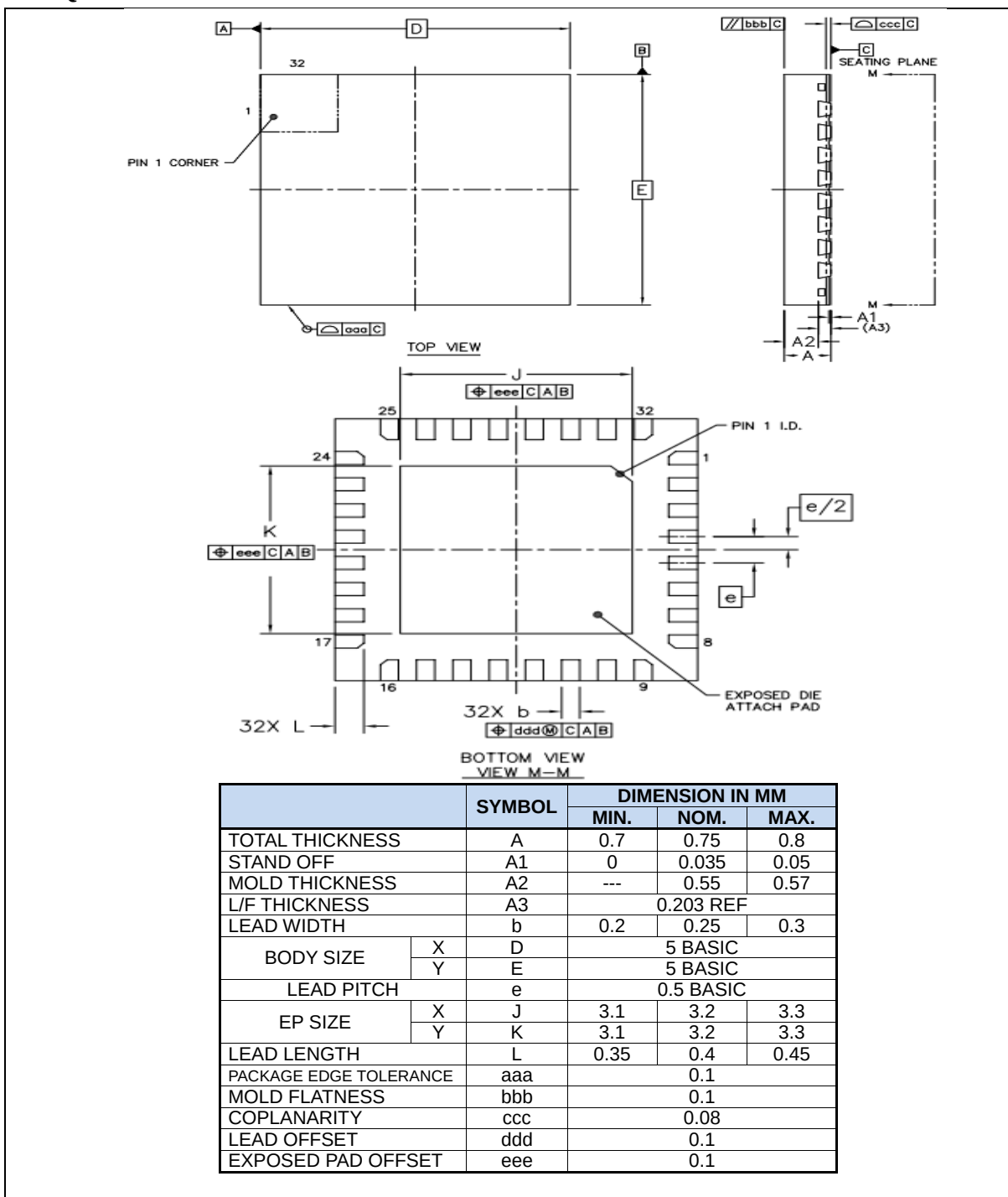
SPI Timing

| Symbol             | Parameter   | Min | Typ. | Max | Units | Notes |
|--------------------|-------------|-----|------|-----|-------|-------|
| SPI <sub>CLK</sub> | SPICLK=0x00 | --  | --   | 16  | MHz   |       |
|                    | SPICLK=0x01 | --  | --   | 8   | MHz   |       |
|                    | SPICLK=0x02 | --  | --   | 4   | MHz   |       |
|                    | SPICLK=0x03 | --  | --   | 2   | MHz   |       |
|                    | SPICLK=0x04 | --  | --   | 1   | MHz   |       |

| Symbol            | Parameter                      | Min.   | Typ. | Max. | Unit | Notes |
|-------------------|--------------------------------|--------|------|------|------|-------|
| t <sub>SCLK</sub> | SPI <sub>CLK</sub> period      | 62.5nS | --   | 1uS  | --   |       |
| t <sub>CLH</sub>  | SPI <sub>CLK</sub> High Period | --     | 50   | --   | %    |       |
| t <sub>CLL</sub>  | SPI <sub>CLK</sub> Low Period  | --     | 50   | --   | %    |       |

## 5. Packaging Information

### 5.1 QFN-32 Dimensions



**QFN32 Dimensions**